

THE NATURE CONSERVANCY OF CANADA DARKWOODS FOREST CARBON PROJECT

Document Prepared By 3GreenTree Ecosystem Services Ltd.

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GHG Accounting/Crediting Period	01 April 2008 – 31 March 2109; 100-year total period
Monitoring Period of this Report	01 January 2013 – 31 December 2016
History of CCB Status	CCB Validation: 30 June 2014
Gold Level Criteria	Biodiversity – The project achieves exceptional biodiversity benefits due to the conservation of critical habitat for endangered ecosystems and species on the project area.

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1 SUMMARY OF PROJECT BENEFITS

The Darkwoods Forest Carbon Project creates climate, community and biodiversity benefits through the acquisition and management of a 54,792 ha (135,394 acre) commercial timberland for ecological conservation purposes in perpetuity. The project avoids GHG emissions from logging in the without-project baseline scenario, retains and sequesters significant carbon stocks, retains substantial net benefits for the local communities, and provides exceptional biodiversity benefits from the conservation and management of large-scale fully functional natural ecosystems and critical habitat for key endangered species.

1.1 Unique Project Benefits

Outcome or Impact	Achievements during the Monitoring Period	Section Reference	Achievements during the Project Lifetime
1) The conservation of the Darkwoods landbase has provided a 'living laboratory' in which scientists have been able to conduct a wide variety of research projects ranging from insect impacts on forest health to wildlife conservation strategies.	16 different research groups were active on the Darkwoods landbase during the monitoring period.	4.3.2	At least 16 different research groups active on the Darkwoods landbase
2) The conservation of the Darkwoods landbase and the managed public access has provided a wide range of unique recreation opportunities for a considerable number of community groups and clubs.	12 different recreation clubs and groups have used the Darkwoods landbase and 488 public use permits have been issued during the monitoring period.	4.3.2	At least 12 different recreation clubs and groups have used the Darkwoods landbase and > 500 public use permits have been issued during the project lifetime.

1.2 Standardized Benefit Metrics

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
GHG emission reductions & removals	Net estimated emission removals in the project area, measured against the without-project scenario	Not Applicable		Not Applicable
	Net estimated emission reductions in the project area, measured against the without-project scenario	1,559,360 tCO ₂ e	3.2.4	3,511,615 tCO ₂ e
Forest ¹ cover	For REDD ² projects: Number of hectares of reduced forest loss in the project area measured against the without-project scenario	Not Applicable		Not Applicable
	For ARR ³ projects: Number of hectares of forest cover increased in the project area measured against the without-project scenario	Not Applicable		Not Applicable
Improved land management	Number of hectares of existing production forest land in which IFM ⁴ practices have occurred as a result of	Net 2,407.4 ha ⁵ through avoided harvesting for the	3.1.3 ⁶	Net 6,082 ha avoided harvest to date.

¹ Land with woody vegetation that meets an internationally accepted definition (e.g., UNFCCC, FAO or IPCC) of what constitutes a forest, which includes threshold parameters, such as minimum forest area, tree height and level of crown cover, and may include mature, secondary, degraded and wetland forests (*VCS Program Definitions*)

² Reduced emissions from deforestation and forest degradation (REDD) - Activities that reduce GHG emissions by slowing or stopping conversion of forests to non-forest land and/or reduce the degradation of forest land where forest biomass is lost (*VCS Program Definitions*)

³ Afforestation, reforestation and revegetation (ARR) - Activities that increase carbon stocks in woody biomass (and in some cases soils) by establishing, increasing and/or restoring vegetative cover through the planting, sowing and/or human-assisted natural regeneration of woody vegetation (*VCS Program Definitions*)

⁴ Improved forest management (IFM) - Activities that change forest management practices and increase carbon stock on forest lands managed for wood products such as saw timber, pulpwood and fuelwood (*VCS Program Definitions*)

⁵ Calculated as (2013-16): Baseline activities = 2,447.0 ha vs. Project activities = 39.6 ha; net = 2,407.4 ha where harvesting was avoided by the project scenario. Project to date values calculated similarly, for the period 2008 to 2016.

⁶ Hectares are not specifically reported in this document and can be found in the ATLAS model output file: 'ATLAS Runs updated Dec 2012.xlsx'

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
	the project's activities, measured against the without-project scenario	project period		
	Number of hectares of non-forest land in which improved land management has occurred as a result of the project's activities, measured against the without-project scenario	Not Applicable		Not Applicable
Training	Total number of community members who have improved skills and/or knowledge resulting from training provided as part of project activities	Data not available		Data not available
	Number of female community members who have improved skills and/or knowledge resulting from training provided as part of project activities	Data not available		Data not available
Employment	Total number of people employed in of project activities, ⁷ expressed as number of full time employees ⁸	57 FTE ⁹	4.1 4.3.2 Table 18 Table 19	292 FTE ¹⁰
	Number of women employed in project activities, expressed as number of full time employees	Data not available		Data not available
Livelihoods	Total number of people with improved livelihoods ¹¹ or income generated as	Data not available		Data not available

⁷ Employed in project activities means people directly working on project activities in return for compensation (financial or otherwise), including employees, contracted workers, sub-contracted workers and community members that are paid to carry out project-related work.

⁸ Full time equivalency is calculated as the total number of hours worked (by full-time, part-time, temporary and/or seasonal staff) divided by the average number of hours worked in full-time jobs within the country, region or economic territory (adapted from UN System of National Accounts (1993) paragraphs 17.14[15.102];[17.28])

⁹ See project monitoring period total results in Table 18 plus results in Table 19.

¹⁰ See CCB PDD (Table 17 and Table 18) for period 2008-2012, added to this monitoring period result.

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
	a result of project activities			
	Number of women with improved livelihoods or income generated as a result of project activities	Data not available		Data not available
Health	Total number of people for whom health services were improved as a result of project activities, measured against the without-project scenario	Not Applicable		Not Applicable
	Number of women for whom health services were improved as a result of project activities, measured against the without-project scenario	Not Applicable		Not Applicable
Education	Total number of people for whom access to, or quality of, education was improved as a result of project activities, measured against the without-project scenario	Not Applicable		Not Applicable
	Number of women and girls for whom access to, or quality of, education was improved as a result of project activities, measured against the without-project scenario	Not Applicable		Not Applicable
Water	Total number of people who experienced increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	Data not available		Data not available
	Number of women who experienced increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	Not Applicable		Not Applicable

¹¹ Livelihoods are the capabilities, assets (including material and social resources) and activities required for a means of living (Krantz, Lasse, 2001. *The Sustainable Livelihood Approach to Poverty Reduction*. SIDA). Livelihood benefits may include benefits reported in the Employment metrics of this table.

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
Well-being	Total number of community members whose well-being ¹² was improved as a result of project activities	Data not available		Data not available
	Number of women whose well-being was improved as a result of project activities	Data not available		Data not available
Biodiversity conservation	Change in the number of hectares significantly better managed by the project for biodiversity conservation, ¹³ measured against the without-project scenario	Data not available		Data not available
	Number of globally Critically Endangered or Endangered species ¹⁴ benefiting from reduced threats as a result of project activities, ¹⁵ measured against the without-project scenario	3 IUCN Red-Listed Species 15 significant species at risk	5.2.2	3 IUCN Red-Listed Species 15 significant species at risk

¹² Well-being is people's experience of the quality of their lives. Well-being benefits may include benefits reported in other metrics of this table (e.g. Training, Employment, Health, Education, Water, etc.), but could also include other benefits such as empowerment of community groups, strengthened legal rights to resources, conservation of access to areas of cultural significance, etc.

¹³ Biodiversity conservation in this context means areas where specific management measures are being implemented as a part of project activities with an objective of enhancing biodiversity conservation.

¹⁴ Per IUCN's Red List of Threatened Species

¹⁵ In the absence of direct population or occupancy measures, measurement of reduced threats may be used as evidence of benefit

2 GENERAL

2.1 Project Description

2.1.1 Implementation Description

The Darkwoods Forest Carbon Project has been fully implemented under NCC ownership and direct management during this monitoring period, which includes years 6,7,8, and 9 of the project. Active management towards the projects conservation objectives are fully funded and ongoing annually on a daily/weekly basis. As planned, the project implementation includes a low level of timber harvesting, forest management and protection activities, various scientific and biodiversity activities, recreational use, and other non-consumptive uses that support the community, biodiversity, and climate goals of the project.

By retaining and conserving the majority of the project area, NCC has continued to provide net ecosystem services benefits (i.e. emissions reductions, water quality, etc.) and key ecological benefits for biodiversity. This includes significant protection and enhancements of threatened and endangered species and habitat, along with large-scale fully functioning ecosystems. The local communities continue to benefit from ecosystem services provisioning, economic benefits, and local amenity improvements.

During this monitoring period, the most significant project activity was the harvest of 17,825 m³ of timber on 39.6 ha over the four years. Five lightning strike fires occurred during the period, impacting 182.8 ha, and the resulting carbon changes have been accounted for in the ex-post calculations. There has been no activity shifting leakage, and market leakage has been tracked and accounted for following the methodology. Significant biodiversity, ecosystem services, and other community net benefits continue to result from the project during this monitoring period. The project continues to manage and mitigate project risks in a manner consistent with the VCS AFOLU Non-Permanence Risk Tool assessments.

The project has generated net GHG emission reductions of 1,559,360 tCO₂e over this four year monitoring period. The VCS Non-Permanence Risk Rating remains unchanged at 10%.

2.1.2 Project Category and Activity Type

Sector 14 - AFOLU

Improved Forest Management (IFM),

Logged Forest to Protected Forest (LtPF).

Not a grouped project.

2.1.3 Project Proponent(s)

Organization name	Nature Conservancy of Canada
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2.1.4 Other Entities Involved in the Project

Organization name	3GreenTree Ecosystem Services Ltd.
Role in the project	Project Developer
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2.1.5 Project Start Date (G3.4)

Project Start Date: April 1, 2008

2.1.6 Project Crediting Period (G3.4)

Project Start Date: 01 April 2008

Project End Date: 31 March 2109

Crediting Period: 100 years

2.1.7 Project Location (G3.3)

The Darkwoods property is a 54,792 ha (135,394 acre) contiguous parcel of fee simple private property in south eastern British Columbia just north east of the municipality of Creston. The Darkwoods property is bounded by Kootenay Lake on the east and various crown and private land on the other property boundaries. There is a significant in-holding in the center of the property that is owned and managed by

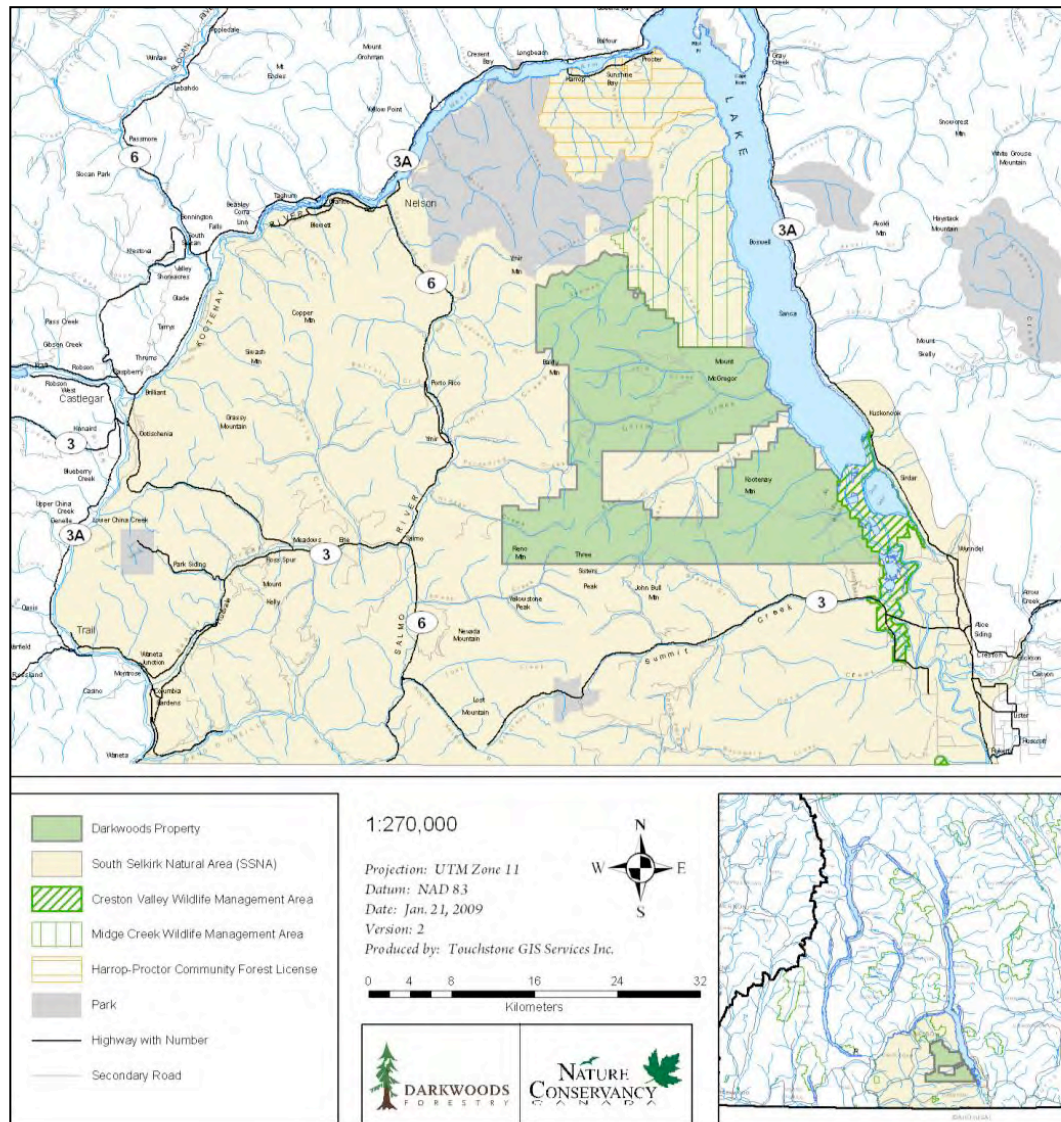
the Windall Box Company, with access rights through the Darkwoods property. The boundaries are surveyed (Table 1) with the following legal and land description:

Table 1. Legal Descriptions and Parcel Identifiers of Darkwoods Parcels

PID #	Description
PID 007-608-349	Sublot 1, District Lot 2381, Kootenay District, Plan X74, except (1) Plans 1760 and NEP77791 and (2) Part in Lot 15184 known as "Tramline" MC
PID 007-608-594	Sublot 7, District Lot 2381, Kootenay District, Plan X74
PID 007-608-446	District Lot 887, Kootenay District, except part included in Plan 1760
PID 026-235-927	District Lot 15184, Kootenay District

There have been no changes to the property boundary or project area during this monitoring period.

Figure 1. Darkwoods Property Overview Map



2.1.8 Title and Reference of Methodology

Methodology: VM0012 - Improved Forest Management in Temperate and Boreal Forests (LTPF), v1.0

Tool Applied: VCS AFOLU Non-Permanence Risk Tool v.3.3

CCB Validation Standard: CCB Standards Edition 2

Validated Project Description Documents:

- VCS PDD: Darkwoods Forest Carbon Project – Version 1.8. Validated April 21, 2011.
- CCB PDD: Darkwoods Forest Carbon Project – Version 1.8. Validated June 30, 2014

2.1.9 Other Programs (CL1.5)

Darkwoods does not participate in any other emission trading, GHG program, sustainable development program, or other social or environmental credit program.

2.1.10 Sustainable Development

Not applicable. Darkwoods does not participate in achieving any nationally stated sustainable development priorities.

2.2 Project Implementation Status

2.2.1 Implementation Schedule (G3.4)

The Darkwoods project has been fully implemented and has consistent and stable ongoing project activities and monitoring. This monitoring period represents years 6, 7, 8 and 9 of the ongoing project and includes the calendar years 2013 – 2016, inclusive. Previously, projects years 1 – 6 (2008-20012) have been successfully verified under VCS. This is the first CCBA project implementation verification.

During this monitoring period, NCC has undertaken low-level timber harvesting in the project scenario as planned, along with planned forest management and protection activities. The monitoring period harvest levels are consistent with the projected ex-ante project scenario activities in the PDD.

The project has undertaken all required monitoring activities on an annual basis.

Date	Milestone(s) in the project's development and implementation
Apr 1, 2008	Property acquisition and project start date.
Apr 21, 2011	VCS Project Validation
Apr 21, 2011	VCS Project Verification, 2008, 2009, 2010
Mar 3, 2014	VCS Project Verification, 2011, 2012
Jun 30, 2014	CCB Project Validation

Annually	Project implementation: ongoing NCC ownership and property management.
Annually	Project area monitoring on a daily/weekly and annual basis

2.2.2 Methodology Deviations

The following methodology deviations have been identified this verification period. Their nature and impact are explained in the following section.

1. A minor methodology deviation has occurred with respect to the timing of plot re-measurement. The methodology requires that all permanent sample plots must be re-measured within a 5-year period. However, there were 16 of the 67 plots that were installed during the Spring of 2011 that were not re-measured until the fall of 2016 and thus exceed the time limit by approximately 6-months. The reason for this deviation is that we did not want to measure those 16 plots during the winter (to meet the 5-year deadline) as the snow cover would have made it impossible to measure downed wood. Further, the cost of measuring the 16 plots separately from the other 51 plots was prohibitive. Thus, the decision was made to measure all 67 together in the fall of 2017. Considering the slow growth of these forest ecosystems it is clear that this 6-month delay in measuring these plots had no impact on the calculation of saleable VCUs determined for the verification period. We can be certain of this statement as the calculation of the uncertainty factor is based on the determination Project Error term (P_E) which is calculated as the sum of two error terms (Model Error (E_M) and Inventory Error (E_I)) that are estimated from the difference of modeled predictions of net ecosystem carbon storage relative to plot-measured net ecosystem carbon storage. Six months of growth would only amount to very small changes in total measured plot biomass relative to the total ecosystem carbon storage in each stand. Since the P_E term was (4.0%) well below the 10% error threshold defined in the methodology, the Uncertainty Factor discount would remain at 1.5%. Lastly, since this deviation is only about the timing of the plot measurements, it clearly only relates to the criteria and procedures for monitoring and measurement and does not relate to any other part of the methodology.
2. There is a minor deviation from the methodology and related monitoring plan elements described in the VCS PDD. The plot network while expanded considerably from that in place during the first monitoring period still does not meet the stated criteria of having a minimum of one plot within each analysis unit. Despite the addition of the 12 new plots in 2014 to resolve a previous monitoring period deviation, there is still one analysis unit without plot representation (AU 110). However, the area in this unrepresented analysis unit represents <0.01% of the total productive forest area (See Table 5) so it has a no impact on the calculation of total VCUs from the project activities. To demonstrate this, a sensitivity analysis was conducted by adding a hypothetical plot for this AU and assigning measured carbon values ranging from the highest value measured in any plot to a value of 0. We then included this hypothetical plot in the calculation of the Uncertainty Factor using the equations built into the 'Uncertainty Factor' worksheet in the 'Darkwoods plot data Jun 2017 v1.5.xlsx' spreadsheet. The results of the sensitivity analysis showed no impact on the value of the uncertainty factor due largely to the very small area represented by AU 110. Lastly, since this deviation only affects the plots used to verify model estimates of net ecosystem carbon storage for specific forest analysis units, it clearly only relates

to the criteria and procedures for monitoring and measurement and does not relate to any other part of the methodology.

3. A minor methodology deviation has occurred with respect equation 60b in the methodology VM0012. Specifically the $y_{m,h,i}$ and $y_{p,h,i}$ terms are reversed in equation 60b (see below). This is a known error with the methodology and it has been acknowledged as such by the project developers and the audit team. Since the equation has been applied correctly in the current verification, there is no negative impact on the conservativeness of the calculation of saleable VCUs. The calculation of the GHG emissions using the corrected version of equation 60b will be different from those that would have been made using the incorrect version of the equation in that the direction of the calculated model error term is negative instead of positive. This is important as the model is underestimating biomass on average relative to the plot data. Lastly, since this deviation is only relevant for the calculation of mean model error in equation 60b, it clearly only relates to the criteria and procedures for monitoring and measurement and does not relate to any other part of the methodology.

From VM0012:

$$y_{d,h,i} = A_{PRJ,h} \cdot (y_{m,h,i} - y_{p,h,i}) \quad (60b)$$

E_M = Mean model error for the project (%)

$y_{d,h,i}$ = the area-weighted difference between measured and predicted carbon storage in analysis unit, h , plot observation, i (t C)

$y_{m,h,i}$ = carbon storage measured in analysis unit, h , plot observation, i (t C ha⁻¹)

$y_{p,h,i}$ = carbon storage predicted by model for analysis unit, h , plot observation, i (t C ha⁻¹)

$A_{PRJ,h}$ = area of project analysis unit, h (ha)

4. A minor methodology deviation has been identified with respect to the decision to stop measuring plots that had been established in the non-productive landbase (where the New C_AU field = '111' in the Darkwoods_VRI_Resultant_Jul2017 v1.1.mdb database). A decision was made to drop plots in the non-productive landbase as they have no impact on the calculation of saleable VCUs. This deviation will have no impact on the calculation of GHG emission reductions for the project because the non-productive landbase is never harvested in the baseline scenario and thus never deviates from the project scenario. Lastly, since this deviation only affects the plots used to verify model estimates of net ecosystem carbon storage for specific forest analysis units, it clearly only relates to the criteria and procedures for monitoring and measurement and does not relate to any other part of the methodology.
5. A minor methodology deviation has been identified with respect to the determination of the amount of harvested wood used in the calculation of storage in harvested wood products. Specifically, we used provincial scaling reports to verify the amount of wood harvested from the Darkwoods landbase as it should be more accurate than model projections. Version 1.0 of the VM0012 methodology does not give clear direction where this value should come from and, as such, it is implied that it should come from the modeled value of stemwood biomass removed

from harvested polygons. However, as clarified in section 8.2.10 of VM0012 V1.2, the proponent is allowed to derive data with respect to the quantity of harvested wood biomass removed from the landbase using other data. Since this section was not included in the original version of the methodology, this must be identified as a methodology deviation. This deviation has no impact on the conservativeness of the calculation of GHG emission reductions as it represents the most accurate data. Lastly, since this deviation only affects the method for the determination of harvested wood biomass in the project scenario, it clearly only relates to the criteria and procedures for monitoring and measurement and does not relate to any other part of the methodology.

2.2.3 Minor Changes to Project Description (Rules 3.5.6)

Not applicable.

2.2.4 Project Description Deviations (Rules 3.5.7 – 3.5.10)

1. A project description deviation has occurred with respect to the spatial representation of project boundary. Upon closer investigation, we determined that there was a small error in the original KML file submitted to VCS. The difference in area was ~ 1.21 ha out of 54,792 ha (< 0.003%) of the total area. Upon advice from the VCS Secretariat, we will report this as a project design deviation and have included a corrected version of the boundary for use in future verifications (KML_607_v1.1.kmz). While the original KML file provided on the VCS website was mistakenly submitted to VCS following the completion of the project validation (April 2011), the correct version of the project boundary was used in the validation modelling work and has been used in all of the modelling in this verification and the last verification (2011-2012) as shown in the forest inventory file 'Darkwoods_VRI_Resultant_Jul2017 v1.1.mdb'. Evidence that this boundary file was used in the validation modelling work and the previous verification (2011-2012) can be observed by viewing the spatial database files included with those submissions. These files are available upon request. This deviation relates to the definition of the project boundary area only. However, as noted above, the correct boundary layer has been used in all calculations conducted in the PD validation and all subsequent monitoring periods. As such this deviation has had no impact on the applicability of the methodology, additionality or the appropriateness of the baseline scenario. Further, the deviation had no impact on the outcomes of any calculations for the reasons stated above.
2. A project description deviation has occurred with respect to the use of literature values of wood density for the different decay classes (DLDWc,i,t) to quantify the mass of lying dead wood in the project scenario as outlined in the monitoring plan described in Section 3.3 of the PD. While somewhat ambiguous, the PD does allow for the use of literature-derived estimates of decay classes and wood density values (see footnote 8 in Section 3.3 of the PD). Further, literature estimates of wood densities for the specific decay classes were used in for calculations in the PD and in the previous monitoring report (2011-2012). However, since the default approach outlined by the methodology prescribes the direct measurement of wood density in the lab, we think a project deviation is warranted. The literature method was selected as it was determined that the direct measurement approach was too time consuming and not cost effective. Further, upon

closer examination, we expected that the direct measurement approach with its limited sample size ($n=10$) would not provide values that were more accurate than literature-derived values with higher sample sizes. Lastly, we conducted a sensitivity analysis (see Darkwoods Plot Data Jun 2017 v1.5, Decay classes sheet) in which we inserted values for wood density for the 3 decay classes that were $\pm 50\%$ from the literature-derived values and showed that they had no impact on the estimation of the uncertainty factor error term. Thus, this deviation does not affect the conservativeness of the GHG emission reduction calculations for the project.

In terms timing, this deviation has been present since project validation and during the previous verification period (2011-2012). It has been identified during the current verification period (2013-2016), but it should be noted that the approach of using literature-derived values to estimate the density of wood in different decay classes has been consistent throughout. Since this approach has been used since project validation and has very little impact on the determination of the uncertainty factor (as shown in the sensitivity analysis above) we can state with confidence that the deviation has had no impact on the applicability of the methodology, additionality or the appropriateness of the baseline scenario. Using the same logic, it is also evident that there has been no impact of this deviation on the estimation of GHG emission reductions for the project.

3. A project description deviation has been identified with respect to the update of the forest inventory that occurred prior to the 2011-2012 Darkwoods verification report. It should have been noted in the previous monitoring report but was not identified as such. Specifically, In 2011, Nature Conservancy Canada engaged Ecora Resource Group Ltd (Ecora) to complete an integrated Vegetation Resource Inventory (VRI) and Terrestrial Ecosystem Mapping (TEM) project on the Darkwoods property. One of the stated objectives was to improve the overall quality of the spatial forest vegetation data in support of ongoing forest resources management. Aerial photography was acquired for the property in July 2011. The product included high-resolution digital imagery (20cm) and a high-resolution ortho-photo. The digital imagery in combination with a field sampling regime were used as the basis to create the VRI data which were used to support the continued monitoring of the Darkwoods forest carbon project. The complete report describing the development of VRI and TEM data is provided as a supporting document entitled: "Vegetation Resource Inventory And Terrestrial Ecosystem Mapping For The Darkwoods Property, British Columbia: Final Report" and dated April, 2012.

As described in detail in the 2011-2012 Darkwoods Monitoring report, all of the assumptions with respect to the pre-stratification of the project landbase (e.g. defining the productive forest landbase and the timber harvesting landbase) were consistent with those utilized for the project validation work with the previous inventory. This process is described in detail in Section 3.2.1. However, the fact that the underlying forest polygons used in the calculation of carbon storage have changed since the project validation required that a project description deviation was identified and issued. The updated inventory, which clearly provides improved accuracy relative to the older inventory used at validation does not negatively influence the conservativeness of the calculation of GHG emission reductions. In contrast, it has improved the accuracy of the dependent carbon calculations. This claim is supported through the error analysis conducted using the data from the sample plot network. Specifically the deviation relates to the fact that the baseline was run with the new inventory, resulting in modifications to data and parameters available at validation. These include different inventory polygons (ABSL,i, APRJ,i) and a slightly

different timber harvesting landbase (THLB). However, the project scenario was run using the same updated inventory. The reason for the changes was to accommodate the improvements in accuracy related to updating the inventory data as described above. Further, since there were no changes with respect to the assumptions that were employed to determine the timber-harvesting landbase, and there were no changes in the underlying assumptions for the management activities employed in the baseline scenario, the deviation has had no impact on the applicability of the methodology, additionality or the appropriateness of the baseline scenario. The outcome of the deviation is that it has an impact on the calculation of emission reductions in the baseline and project scenarios related to the underlying inventory polygons. While it is not possible to specifically show the impact of this deviation, it has resulted in a more accurate estimate of project-related GHG emission reductions. Specifically, the parameters describing the project & baseline inventory polygons in Section 3.1 of the PD have been updated. These updates also led to changes in the emission reductions calculations made for the baseline scenario in Section 4.1 of the PD.

4. A minor projection description deviation has been noted related to the methodology deviation (MD) #2 described in Section 2.2.2. Specifically, the PD suggested that the plot network would be finalized by 2011. As stated in MD#2, there was an addition of 12 new plots in 2014 to address the fact that there were analysis units without plot representation following the 2011-2012 verification. Moreover, there is still one analysis unit without plot representation (AU 110) although it is very small and has no impact on the calculation of the uncertainty factor (see Section 2.2.2, deviation #2). We expect to drop this analysis unit for the next verification period. This deviation occurred in the year following the 2011 target date suggested in the PD. The outcome of this deviation is negligible for the reasons stated above (and see Section 2.2.2, deviation #2). Further, the deviation has had no impact on the applicability of the methodology, additionality or the appropriateness of the baseline scenario. Specifically, the number of analysis units used in the calculation of the uncertainty factor (Section 4.5 of the PD) and the statement that the plot network would be finalized by 2011 (Section 3.3 of the PD) have been updated..

5. A minor project description deviation has been noted related to the methodology deviation (MD) #1 described in Section 2.2.2. Specifically, the methodology and PD state that “all permanent sample plots must be re-measured within a 5-year period”. However, there were 16 of the 67 plots that were installed during the Spring of 2011 that were not re-measured until the fall of 2016 and thus exceed the time limit by approximately 6-months. The reason for this deviation is that we did not want to measure those 16 plots during the winter (to meet the 5-year deadline) as the snow cover would have made it impossible to measure downed wood. Further, the cost of measuring the 16 plots separately from the other 51 plots was prohibitive. As explained in detail in Section 2.2.2, MD #1, this deviation has no impact on the determination of GHG emission reductions. This deviation occurred in the fall of 2016 when the plot re-measurements were conducted for current verification period. The outcome of this deviation is negligible for the reasons stated above (and see Section 2.2.2, deviation #1). Further, the deviation has had no impact on the applicability of the methodology, additionality or the appropriateness of the baseline scenario. Specifically, Section 3.3 of the PD has been updated to accommodate a 5.5 yr length of time between measurements for 16 of 67 sample plots.

2.2.5 Risks to the Project (G3.5)

During this monitoring period, the project was subject primarily to the risks of fire, pests and disease, and extreme weather. The project area is access controlled (which mitigates and limits any potential human-induced risks), and local NCC staff are regularly on the property to monitor project activities and natural disturbance conditions. In addition to observation by NCC staff, contractors, and other users of the property during day-to-day activities, NCC undertakes at least one observation helicopter flight over the property per year to survey for unexpected impacts from pests, disease, extreme weather, and/or fire. Additional over flights may be undertaken seasonally, or in reaction to heavy storms, high fire conditions, or other reasons.

During daily and helicopter observations during this monitoring period, no material areas of pest or disease beyond normal endemic levels were identified. Also, no material areas of forest damage due to extreme weather was noted beyond normal and expected levels of individual tree blowdown. The project area was subject to several (2 non-de minimis) natural forest fires during this monitoring period, which is consistent with the risk assessment outlined in the VCS AFOLU Non-Permanence Risk Tool. The project's fire detection and fire-fighting processes worked successfully as expected. As outlined in the VCS PDD, NCC's fire protection agreement with the BC Ministry of Forests provides primary fire risk mitigation from advanced automated provincial fire detection processes, along with initial attack capacity located immediate adjacent to the Darkwoods property). As expected, the BC Ministry of Forests detected and undertook initial attack and firefighting actions that controlled and limited the impact of fires to project carbon emissions. The net emissions related to these fires were included in annual emission reduction calculations this monitoring report without an unplanned loss event.

No other risks to the project occurred during this monitoring period.

2.2.6 Enhancement of High Conservation Values (G3.6)

High conservation values are protected by implementation of the project. No additional steps were necessary to protect high conservation values during this monitoring period.

2.2.7 Benefit Permanence (G3.7)

The Darkwoods project is based around the conservation of the property in perpetuity and no additional measures were necessary to maintain or enhance the climate, community, and biodiversity benefits during this monitoring period.

2.3 Stakeholder Engagement

2.3.1 Community Consultation (G3.8)

NCC has an open and constant engagement policy with interested stakeholders related to the Darkwoods project and other NCC initiatives in the region. NCC is publically known throughout the region, and any person wanting to comment can do so through phone, social media, website contacts, or direct interaction at the local NCC offices.

During this monitoring period, NCC maintained webpages specific to Darkwoods and the Darkwoods Carbon Project (http://www.natureconservancy.ca/en/where-we-work/british-columbia/featured-projects/darkwoods/dw_carbon.html).

NCC also met with community members and project stakeholders in multiple meetings during this monitoring period. A non-exhaustive list is shown in Section 5.3.3, and example of actively engaged community organizations is shown in Table 23 and Table 24.

These meetings have influenced the project area in multiple ways, from reviewing and developing recreational access agreements; to project area research planning and review; to general public outreach and education.

2.3.2 Public Comment Period Publicity (G3.9)

Stakeholders in the project zone are fully modern with virtually the entire population with daily online access and extensive usage of internet resources. Therefore, NCC has placed a prominent posting on the Darkwoods website with a link to the VCS/CCBA comment website.

2.3.3 Distribution of Project Information (G3.9)

Darkwoods is located in a fully developed region with almost ubiquitous daily internet access, and therefore the primary form of project information distribution is through the NCC project webpages. Project implementation and project area activities are also communicated through meetings with key stakeholders. A non-exhaustive list of stakeholder meetings during the monitoring period is shown in Section 5.3.3. NCC has not identified particular newly affected stakeholders to whom documentation is to be directly distributed; however, NCC is willing to provide documentation should any stakeholders request it.

2.3.4 Conflicts and Grievances (G3.10)

NCC receives feedback generally about regional initiatives and activities on an ongoing basis, and follows the procedures outlined in the CCB PDD. No specific conflicts or grievances related to Darkwoods project implementation have been noted during this monitoring period.

2.4 Management Capacity and Best Practices

2.4.1 Required Technical Skills and Expertise (G4.2)

NCC has retained or improved its regional and national expertise in biodiversity management, carbon measurement and monitoring, and community engagement throughout this monitoring period.

Contact	Company	Title	Project Responsibilities
Michael Bradstreet (MB)	Nature Conservancy of Canada	SVP, New Conservation Strategies. National Office	Overall executive responsibility for carbon finance within NCC.
Tom Swann (TS)	Nature Conservancy	BC Assoc. Regional VP & Director of Land	Overall responsibility for Darkwoods, NCC carbon lead team.

	of Canada	Securement & Major Policy Initiatives	
Rob Wilson (RW)	Nature Conservancy of Canada	Director, Conservation Finance	Carbon project oversight, carbon sales and marketing, NCC carbon lead team.
Nancy Newhouse, RPBio (NN)	Nature Conservancy of Canada	Program Manager, Canadian Rocky Mountains Program, BC Region	Regional manager with direct oversight of Darkwoods operations, regional stakeholder engagement, etc. Project monitoring activities.
Adrian Leslie, RPBio (AL)	Nature Conservancy of Canada	South Selkirk Project Manager	Darkwoods operational management, project monitoring activities.
Jane Gilbert (JG)	Nature Conservancy of Canada	Chief Communications Officer	NCC national communications.
Mike Vitt, BScFor, MBA (MV)	3GreenTree Ecosystem Services Ltd.	Managing Director	Carbon project development, monitoring management.
Clive Welham, PhD, RPBio (CW)	3GreenTree Ecosystem Services Ltd.	Managing Director	Carbon project development, monitoring management.
Brad Seely, PhD (BS)	3GreenTree Ecosystem Services Ltd.	Managing Director	Carbon project development, carbon modeling, monitoring management.

2.4.2 Worker Training (G4.3)

NCC outlines orientation and training processes in: The Nature Conservancy of Canada Management Policy: Human Resources Policy and the NCC Darkwoods Occupational Health & Safety Program.

Other specific technical tasks such as carbon monitoring plot installation and measurement are trained at the start of each field season using standard operating procedure documentation.

2.4.3 Community Employment Opportunities (G4.4)

NCC's employment standards are documented in: The Nature Conservancy of Canada Management Policy: Human Resources Policy. NCC is compliant with all federal and provincial legislation in regards to both employment equity and human rights more broadly. NCC employs locally when feasible. For example, a local person staffs the Darkwoods manager position.

2.4.4 Relevant Laws and Regulations Related to Worker's Rights (G4.5)

Canada has very well developed labour legislation governing workers' rights. Responsibility for the establishment of labour legislation generally falls under provincial jurisdiction, along with federal authority in certain industries. This legislation governs minimum employment standards that by law define and guarantee workers' rights within the workplace. Each province, as well as the federal government, has legislation governing workplace health and safety, as all Canadian workers have the right to work in a healthy and safe environment. Human rights provisions, fair employment practices and equal pay and anti-discrimination laws are covered under federal and provincial human rights and labour legislation.

In British Columbia, the B.C. Ministry of Labour, Employment Standards Branch is responsible for the administration of the provincial Employment Standards Act. This Act covers the following topic areas: Minimum Wage, Minimum Daily Pay, Meal Breaks, Paydays and Payroll Records, Overtime, Averaging Agreements, Uniforms & Special Clothing, Deductions, Statutory Holidays, Employing Young People Under 15, Compensation for Length of Service, Annual Vacation, Vacation Pay, Leave From Work, Temporary Foreign Workers, Collective Agreements and Resolving Disputes.

The Nature Conservancy of Canada has well-developed internal human resources policies governing the management of NCC's employees. The overarching Human Resources Policy is reviewed annually by senior NCC management and updated when deemed necessary. All NCC employees are informed of the existence of this policy on NCC's intranet website, which addresses the standards, requirements and procedures governing an employee's employment by NCC. All new employees are required to sign a disclosure indicating agreement and adherence to these policies and all NCC employees must renew this agreement on an annual basis.

Subcontractors, defined as service providers who bill NCC for their services, must sign an Agreement for Services contract, the standard template of which includes a separate provision entitled Compliance with Laws and Policies, as follows: "The Contractor shall be knowledgeable of and comply with the provisions of all applicable legislation, By-Laws of local authorities and Regulations in the Province of British Columbia and shall specifically ensure that the Contractor and his/her/its employees are knowledgeable of and performs all obligations imposed by the *Occupational Health and Safety Act* or equivalent in the Province of British Columbia".

In summary, NCC is a national organization fully compliant with the laws and regulations of BC and Canada. NCC employment policies and worker guidelines are documented in the latest versions of: The Nature Conservancy of Canada Management Policy: Human Resources Policy and the NCC Darkwoods Occupational Health & Safety Program.

2.4.5 Occupational Safety Assessment (G4.6)

Darkwoods project activities are typical operations for forestry operations and timberland in BC, and include typical forest operations (logging, road work, etc.) and various forest field work. There are no unusual or unique activities, and therefore risks are typical of operations undertaken commonly across the forest industry.

In Canada, all 3rd party contractors operate as "independent contractors", a specific legal term that requires the contractors to operate and direct their delivery of services and employees independently. Therefore, operational safety procedures are the purview of each contractor. The WorkSafe BC regulations and BC Forest Safety Council SAFE Company programs are the primary methods in BC to assure worker safety in the forest industry, and include training and procedural materials used across the industry. Various levels of performance are self-audited by these programs.

NCC's policies related to health and safety, including safe operating procedures, are fully documented in: NCC Darkwoods Occupational Health & Safety Program. As per this document and other NCC annual plans, employees undergo various health and safety awareness and training programs, as applicable to their job function.

NCC contracts mandate that all contractors have a valid and active WorkSafe BC number. The contractor is then responsible to WorkSafe BC (<https://www.worksafebc.com/en/about-us/who-we-are/mission-vision-values>) to ensure that legal regulations are followed. Logging contracts also specify contractors are certified as BC Forest Safety Council SAFE Company.

NCC undertakes pre-work reviews with contractors as described in the NCC Darkwoods Occupational Health & Safety Program, which includes a pre-work safety site orientation checklist. NCC also ensures that logging contractors and other large contractors have at least equivalent procedures. Researchers are made aware of safety issues through both a research permit and waiver. And finally, general public access permits are also advised of the risks on the permits document.

For this monitoring period, no material new or unique employee or contractor worker health and safety risks were created, and the existing NCC, government, and industry policies were implemented as designed to inform workers and mitigate safety risks as per industry best practices.

2.4.6 Financial Health of Implementing Organization(s) (G4.7)

The Nature Conservancy of Canada is Canada's largest not-for-profit land trust organization with over a 50-year history of financial stability. NCC's financial statements are available at: www.natureconservancy.ca/en/who-we-are/annual-reports.

NCC has remained financially stable during this monitoring period, and the Darkwoods project has been fully funded.

2.5 Legal Status and Property Rights

2.5.1 National and Local Laws (G5.1)

Not applicable - no new laws relevant to the project have been implemented in this monitoring period.

2.5.2 Free, Prior and Informed Consent (G5.3)

Not applicable - as fee simple private land, the registered title documents accord all aspects related to surface rights to NCC. The project area does not contain or involve any areas not privately owned by NCC.

2.5.3 Property Rights Protection (G5.4)

Not applicable - the Darkwoods property is private forestland and does not include any permanent habitation or community activities, and as such the project does not require any relocation or exclusion of communities.

2.5.4 Identification of Illegal Activity (G5.5)

Not applicable - no known illegal activities have occurred on the project area during this monitoring period.

3 CLIMATE

3.1 Monitoring GHG Emission Reductions and Removals

3.1.1 Data and Parameters Available at Validation

These values were determined at the time of validation approved by the auditing team at the time. Brad Seely is responsible for all parameter values in Section 3.1.1 and 3.1.2. Data and parameters monitored during the operation of the project are included in Section 3.1.2 (Data and Parameters Monitored) below.

Data / Parameter	THLB
Data unit	Ha
Description	Timber harvesting landbase area
Source of data	NCC Darkwoods GIS spatial inventory data used for the validation forest_cover2010v3.2. mdb
Value applied	28,549.99 ha
Justification of choice of data or description of measurement methods and procedures applied	The method for identifying the THLB is justified as it follows common practices employed in British Columbia for identifying forested areas in which harvesting is both practice and economical. ("Inventory Data Collection Standards for VRI Ground Sampling Version 2.1" which can be found at: https://www.for.gov.bc.ca/hts/risc/alphastand.htm) The specific criteria used for identifying the THLB is described in Section 3.2.1
Purpose of the data	Required for calculation of baseline and project emissions
Comments	The updated THLB area from the new inventory is 29,497.04 ha as shown in the summary sheet of the Project

Data / Parameter	ABSL_i, APRJ_i
Data unit	Ha
Description	Respective areas of baseline and project polygon, i
Source of data	NCC Darkwoods GIS spatial inventory data used for the validation forest_cover2010v3.2. mdb. This dataset was derived from the inventory data developed over several years for managing the Darkwoods property prior to the purchase of the property by NCC.
Value applied	See original spatial data
Justification of choice of data or description of measurement methods and procedures applied	The method of mapping inventory polygons, identified as homogenous forest management units, is justified as it is based on standard British Columbia inventory methods as outlined in "Inventory Data Collection Standards for VRI Ground Sampling

	Version 2.1" which can be found at: https://www.for.gov.bc.ca/hts/risc/alphastand.htm
Purpose of the data	Required for calculation of baseline and project emissions
Comments	Current values can be seen in DARKWOODS_VRI_JUL2017_DistUpdate table in Darkwoods_VRI_Resultant_Jul2017 v1.1.mdb. All polygons (i) are included in both parameters.

Data / Parameter	CF
Data unit	t C t ⁻¹ d.m.
Description	Carbon fraction of dry matter
Source of data	IPCC 2006
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of the data	Required for calculation of baseline and project emissions
Comments	

Data / Parameter	R _i
Data unit	unitless
Description	Timber harvesting landbase area
Source of data	Root:shoot ratio in polygon, i
Value applied	Based on Li et al. 2003 but modified according to tree age according to Lehtonen et al. 2004. Conifers range from (0.25 to 0.19) as a function of age (5 to 300y); Hardwoods range from (0.24 to 0.18) as a function of age from (5 to 200y).
Justification of choice of data or description of measurement methods and procedures applied	Root biomass is difficult to measure directly. Estimates came from peer-reviewed literature adjusted to account for the common observation that young trees have a relatively higher proportion of biomass belowground compared to older trees. Additional adjustments were made to account for the differences between conifers and hardwoods.
Purpose of the data	Required for calculation of baseline and project emissions
Comments	

Data / Parameter	$f_{BSL,NATURAL,i,t}$, $f_{PRJ,NATURAL,i,t}$
Data unit	unitless ($0 < f_{BSL,NATURAL,i,t}$, $f_{PRJ,NATURAL,i,t} < 1$)
Description	The proportion of biomass that dies from natural mortality in polygon, i, year, t, in the baseline and project cases, respectively.
Source of data	Estimate established over years of FORECAST development comparing model outputs of coarse woody debris and snag accumulation against field data.
Value applied	0.2 % per annum
Justification of choice of data or description of measurement methods and procedures applied	The estimate of the parameter is justified as it is based on many observations (by the experienced modelling team – Brad Seely and Clive Welham) of within stand mortality that is unrelated to stand-self thinning. The estimate has been found to be reasonable when comparing growth and yield projections in the model to field measured values in similar stand types through this southern interior region of BC. It has also been verified in consultation with local experts.
Purpose of the data	Required for calculation of baseline and project emissions
Comments	

Data / Parameter	$f_{BSL,HARVEST,i,t}$, $f_{PRJ,HARVEST,i,t}$
Data unit	unitless ($0 < f_{BSL,HARVEST,i,t}$, $f_{PRJ,HARVEST,i,t} < 1$)
Description	The proportion of biomass removed by harvesting from polygon, i, in year, t, in the baseline and project cases, respectively.
Source of data	Estimate established over years of FORECAST development comparing model outputs of coarse woody debris and snag accumulation against field data.
Value applied	Variable, depends on simulated harvest schedule in the baseline scenario and ex-post harvest in the project scenario (see Project Scenario Projection v1.1.xlsx). In the FORECAST model the user provides stump height and top diameter limits. These values are used to determine the proportion removed.
Justification of choice of data or description of measurement methods and procedures applied	The estimate is justified as it is based on standard stump heights (40cm) and top diameter limits (10cm to 12.5 cm) for different species within the region.
Purpose of the data	Required for calculation of baseline and project emissions
Comments	

Data / Parameter	$f_{BSL,DAMAGE,i,t}$, $f_{PRJ,DAMAGE,i,t}$
Data unit	unitless ($0 < f_{BSL,DAMAGE,i,t}$, $f_{PRJ,DAMAGE,i,t} < 1$)
Description	The proportion of additional biomass removed by for road and landing construction in polygon, i, year, t, in the baseline and project cases, respectively.
Source of data	Determined as part of the inventory work (see Darkwoods Inventory Update Report Darkwoods_VRIEM_Ecora2012.pdf) Monitoring data on an ex-post basis.
Value applied	0 in ex-ante baseline and project scenarios. From monitoring data for project ex-post calculations. Road widths are determined assuming a 5m buffer on either side of the centerline. No new roads have been built since project establishment.
Justification of choice of data or description of measurement methods and procedures applied	Original roads were identified from the photo interpretation work done to create the inventory data. Road widths are determined assuming a 5m buffer on either side of the centerline. The selected value is justified as it represents the standard road width used for forest management planning in BC. No new roads have been built since project establishment.
Purpose of the data	Required for calculation of baseline and project emissions
Comments	

Data / Parameter	$f_{BSL,BLOWDOWN,i,t}$, $f_{PRJ,BLOWDOWN,i,t}$
Data unit	unitless ($0 < f_{BSL,BLOWDOWN,i,t}$, $f_{PRJ,BLOWDOWN,i,t} < 1$)
Description	The proportion of live aboveground tree biomass subject to blowdown in polygon, i, year, t, in the baseline and project cases, respectively.
Source of data	Included within the natural mortality factor calculated in $f_{BSL,NATURAL,i,t}$, $f_{PRJ,NATURAL,i,t}$ Also captured by spatial monitoring if >4ha, which would be incorporated as a new polygon on an ex-post.
Value applied	Zero for the baseline and project ex-ante calculations (part of the natural mortality factor source data).
Justification of choice of data or description of measurement methods and procedures applied	The value (0) selected for the baseline scenario is justified as it represents the most conservative case. The value for the project scenario (0) was determined through ongoing monitoring activities (in the case of large blow down events (>4ha). Adrian Leslie of NCC is currently responsible for identifying and mapping these events in the project area. Prior to 2014 it was Nancy Newhouse of NCC. No dates are provided as there were no blow-down events to

	measure during the project validation period (2008-2010).
Purpose of the data	Required for calculation of baseline and project emissions
Comments	

Data / Parameter	$f_{BSL,BRANCH,i,t}$, $f_{PRJ,BRANCH,i,t}$
Data unit	unitless ($0 < f_{BSL,BRANCH,i,t}$, $f_{PRJ,BRANCH,i,t} < 1$)
Description	The proportion of aboveground tree biomass comprised of branches > 5 cm diameter in polygon, i, year, t, in the baseline and project cases, respectively.
Source of data	Calculated within FORECAST using calibration data from allometric biomass equations by species based upon (Standish, Manning, & Demaerschalk, 1985).
Value applied	Variable, see FORECAST Calibration data sets provided with PD
Justification of choice of data or description of measurement methods and procedures applied	The method for estimating the proportion of branches is justified as it is derived from the most commonly used allometric biomass equations in British Columbia based on the destructive sampling of thousands of trees across 22 species (Standish et al. 1985.)
Purpose of the data	Required for calculation of baseline and project emissions
Comments	

Data / Parameter	$f_{BSL,BUCKINGLOSS,i,t}$, $f_{PRJ,BUCKINGLOSS,i,t}$
Data unit	unitless ($0 < f_{BSL,BUCKINGLOSS,i,t}$, $f_{PRJ,BUCKINGLOSS,i,t} < 1$)
Description	The proportion of the log bole biomass left on site after assessing and/or merchandizing the log bole for quality, in polygon, i, year, t, in the baseline and project cases, respectively.
Source of data	Based on (Smith, Miles, Vissage, & Pugh, 2004), and expert opinion based on FORECAST modeler previous experience.
Value applied	0.10 of stemwood and bark is assumed to be left on site.
Justification of choice of data or description of measurement methods and procedures applied	The estimate is justified as it is based on top diameter limits (see above) and has been verified through consultation with local experts and long-term experience of modelling team.
Purpose of the data	Required for calculation of baseline and project emissions
Comments	

Data / Parameter	$f_{BSL,SNAGFALLDOWN,i,t}$, $f_{PRJ,SNAGFALLDOWN,i,t}$
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Data unit	unitless ($0 < f_{BSL,SNAGFALLDOWNi}, f_{PRJ,SNAGFALLDOWN,i,t} < 1$)
Description	The proportion of snag biomass in polygon, i, year, t, that falls over, in the baseline and project cases, respectively.
Source of data	From: (Parish, Antos, Ott, & Di Lucca, 2010)
Value applied	Variable, depending on species and dbh. Modeled by species and dbh class within FORECAST. FORECAST Calibration data sets provided with PD
Justification of choice of data or description of measurement methods and procedures applied	The estimate is justified as it is based on published snagfall rates in the region (Parish, Antos, Ott, & Di Lucca, 2010). Further it has been verified through consultation with local experts and long-term experience of modelling team.
Purpose of the data	Required for calculation of baseline and project emissions
Comments	

Data / Parameter	$f_{BSL,LWDECAY,i,t}, f_{PRJ,LWDECAY,i,t}$
Data unit	unitless ($0 < f_{BSL,LWDECAYi}, f_{PRJ,LWDECAY,i,t} < 1$)
Description	The annual proportional loss of lying dead biomass due to decay, in polygon i, year, t in the baseline and project cases, respectively.
Source of data	Based upon: (Harmon, et al., 1986), (Laiho & and Prescott, 2004).
Value applied	Variable, modeled within FORECAST, based upon a an exponential decay function similar to: Mass loss occurs in proportion to the amount of mass remaining in accordance with an a single exponential model, of the general form: $Y_t = Y_o e^{-kt}$ where Y_o is the initial quantity of material, Y_t the amount left at time t, and k is a decay constant. FORECAST Calibration data sets provided with PD.
Justification of choice of data or description of measurement methods and procedures applied	The estimate is justified as it is based on published k-values for the species present in the Darkwoods project area derived from references provided above.
Purpose of the data	Required for calculation of baseline and project emissions
Comments	

Data / Parameter	$f_{BSL,SWDECAY,i,t}, f_{PRJ,SWDECAY,i,t}$
Data unit	unitless ($0 < f_{BSL,SWDECAYi}, f_{PRJ,SWDECAY,i,t} < 1$)

Description	The proportional loss of snag biomass due to decay, in polygon, i , year, t , in the baseline and project cases, respectively.
Source of data	Based upon: (Vanderwel, Caspersen, & Woods, 2006a); (Vanderwel, Malcolm, & Smith, 2006b); (Kurz & et al, 2009)
Value applied	Modeled within FORECAST by species based on calibration from the source data references above.
Justification of choice of data or description of measurement methods and procedures applied	As with lying dead wood (see $f_{BSL,LWDECAY,i,t}$), $f_{BSL,SWDECAY,i,t}$ is assumed to occur in proportion to the amount of mass remaining in accordance with a first order exponential model. This estimate is justified as it is based on several peer-reviewed studies of snag and downed-wood decay rates (see above).
Purpose of the data	Required for calculation of baseline and project emissions
Comments	

Data / Parameter	$f_{BSL,DBGDECAY,i,t}$, $f_{PRJ,DBGDECAY,i,t}$
Data unit	unitless ($0 < f_{BSL,DBGDECAY,i,t}$, $f_{PRJ,DBGDECAY,i,t} < 1$)
Description	The proportional loss of dead belowground biomass due to decay, in polygon i , year, t , in the baseline and project cases, respectively.
Source of data	Based upon: (Moore, Trofymow, Siltanen, Prescott, & CIDET, 2005); (Melin, Petersson, & Nordfjell, 2009)
Value applied	Modeled within FORECAST by species based on calibration from the source data references above. See FORECAST Calibration data sets provided with PD.
Justification of choice of data or description of measurement methods and procedures applied	As with lying dead wood (see $f_{BSL,LWDECAY,i,t}$), $f_{BSL,SWDECAY,i,t}$ is assumed to occur in proportion to the amount of mass remaining in accordance with a first order exponential model. This estimate is justified as it is based on several peer-reviewed studies of snag and downed-wood decay rates (see above).
Purpose of the data	Required for calculation of baseline and project emissions
Comments	

Data / Parameter	$f_{BSL,PRODUCTk}$, $f_{BSL,PROCESSk}$, $f_{PRJ,PRODUCTk}$, and $f_{PRJ,PROCESSk}$
Data unit	unitless; $0 < f_{BSL,PRODUCTk}$, $f_{BSL,PROCESSk}$, $f_{PRJ,PRODUCTk}$, and $f_{PRJ,PROCESSk} < 1$
Description	The respective fractions of harvested biomass allocated to a given forest product type, k , and its associated processing efficiency for the baseline (BSL) and project (PRJ) cases.
Source of data	(Thrower & Orr-Ewing, 2004), (Miner, 2006).
Value applied	See Appendix 2 in the Darkwoods Project Design Document

Justification of choice of data or description of measurement methods and procedures applied	This estimate is justified as it was derived from peer-reviewed studies examining the use and processing efficiency of harvested wood in western Canada and the United States.
Purpose of the data	Required for calculation of baseline and project emissions
Comments	

Data / Parameter	$f_{BSL,PERMHWPK,i,t}$, $f_{PRJ,PERMHWPK,i,t}$
Data unit	unitless ($0 < f_{BSL,PERMHWPK,i,t}$, $f_{PRJ,PERMHWPK,i,t} < 1$)
Description	The fraction of biomass allocated to permanent storage, for each product type, k, in the baseline and project cases, respectively.
Source of data	Permanent carbon storage was calculated here using the 100-year method developed by (Miner, 2006).
Value applied	Values are product-specific, as derived below. See Appendix 2, Table 1 in the PD for HL_k values.
Justification of choice of data or description of measurement methods and procedures applied	The method of calculation of $f_{BSL,PERMHWPK}$ (see below) is justified as it is derived from the VCS-accepted 100-year method for identifying storage in harvested wood products developed by Miner (2006) and specified in the VM0012 methodology v1.0. $f_{BSL,PERMHWPK} = (1/(1 + (Ln(2)/HL_k)))^Y$ where: HL_k is the half-life of a given product type, k (years), and Y is the elapsed time (i.e, 100 years).
Purpose of the data	Required for calculation of baseline and project emissions
Comments	

Data / Parameter	$f_{BSL,BARK}$, $f_{BSL,COARSE}$, and $f_{BSL,FINE}$ $f_{PRJ,BARK}$, $f_{PRJ,COARSE}$, and $f_{PRJ,FINE}$
Data unit	unitless ($0 < f_{BSL,BARK}$, $f_{BSL,COARSE}$, $f_{BSL,FINE}$, $f_{PRJ,BARK}$, $f_{PRJ,COARSE}$, $f_{PRJ,FINE} < 1$)
Description	The proportions of bark, coarse, and fine residual biomass, respectively that comprise $B_{BSL,RESIDUAL,t}$ and $B_{PRJ,RESIDUAL,t}$ for the baseline (BSL) and project (PRJ) cases.
Source of data	(Perlack, Wright, Turhollow, Graham, Stodkes, & Erback, 2005)
Value applied	26.5%, for $f_{BSL,BARK}$ and $f_{PRJ,BARK}$ 42.9%, for $f_{BSL,COARSE}$ and $f_{PRJ,COARSE}$ 30.7%, for $f_{BSL,FINE}$ and $f_{PRJ,FINE}$

Justification of choice of data or description of measurement methods and procedures applied	The estimates of these parameters are justified as they are derived from a relevant peer-reviewed study focused on the use of biomass as feedstock for a bioenergy and bioproducts industry.
Purpose of the data	Required for calculation of baseline and project emissions
Comments	

Data / Parameter	$f_{BSL,BARKUSE}$, $f_{BSL,COARSEUSE}$, and $f_{BSL,FINEUSE}$ $f_{PRJ,BARKUSE}$, $f_{PRJ,COARSEUSE}$, and $f_{PRJ,FINEUSE}$
Data unit	unitless ($0 < f_{BSL,BARKUSE}$, $f_{BSL,COARSEUSE}$, $f_{BSL,FINEUSE}$, $f_{PRJ,BARKUSE}$, $f_{PRJ,COARSEUSE}$, $f_{PRJ,FINEUSE} < 1$)
Description	The proportions of bark, coarse, and fine residual biomass, respectively, allocated to secondary manufacturing, for the baseline (BSL) and project (PRJ) cases.
Source of data	(Perlack, Wright, Turhollow, Graham, Stodkes, & Erback, 2005)
Value applied	100%, for $f_{BSL,BARKUSE}$ and $f_{PRJ,BARKUSE}$ 85%, for $f_{BSL,COARSEUSE}$ and $f_{PRJ,COARSEUSE}$ 42%, for $f_{BSL,FINEUSE}$ and $f_{PRJ,FINEUSE}$
Justification of choice of data or description of measurement methods and procedures applied	The estimates of these parameters are justified as they are derived from a relevant peer-reviewed study focused on the use of biomass as feedstock for a bioenergy and bioproducts industry. Evidence indicates that on average 80% of bark is combusted for energy, with the remainder used principally as mulch (Perlack et al. 2005). Decay rates for mulch are difficult to estimate. Hence, as a default, all bark ($f_{BSL,BARKUSE}$) is assumed to be 100% combusted, a conservative assumption.
Purpose of the data	Required for calculation of baseline and project emissions
Comments	

Data / Parameter	$f_{BSL,PROCESSc}$ and $f_{BSL,PROCESSf}$ $f_{PRJ,PROCESSc}$ and $f_{PRJ,PROCESSf}$
Data unit	unitless ($0 < f_{BSL,PROCESSc}$, $f_{BSL,PROCESSf}$, $f_{PRJ,PROCESSc}$, $f_{PRJ,PROCESSf} < 1$)
Description	Processing efficiencies of coarse and fine residuals, respectively, in secondary manufacturing, for the baseline (BSL) and project (PRJ) cases.
Source of data	(Perlack, Wright, Turhollow, Graham, Stodkes, & Erback, 2005)
Value applied	85% to all processing efficiencies

Justification of choice of data or description of measurement methods and procedures applied	The estimates of these parameters are justified as they are derived from a relevant peer-reviewed study focused on the use of biomass as feedstock for a bioenergy and bioproducts industry. The authors note that processing efficiencies of coarse and fine residuals in secondary manufacturing are typically much higher than primary manufacturing.
Purpose of the data	Required for calculation of baseline and project emissions
Comments	

Data / Parameter	BEF
Data unit	unitless
Description	Biomass expansion factors
Source of data	Not applicable
Value applied	No specific BEF are used other than the root:shoot variable described above.
Justification of choice of data or description of measurement methods and procedures applied	
Purpose of the data	
Comments	

Data / Parameter	Allometric equation parameters
Data unit	Unitless
Description	Convert height and DBH into biomass of component pools.
Source of data	Allometric equations from (Standish, Manning, & Demaerschalk, 1985) are used to calibrate biomass modeling within FORECAST.
Value applied	see 'Standish Biomass Equation' worksheet in 'Darkwoods plot data Jun 2017 v1.5.xlsx' These remain unchanged from the original version used in the PD.
Justification of choice of data or description of measurement methods and procedures applied	Used to derive biomass estimates for pools that are difficult to measure. The parameter values are justified because the Standish reference is widely accepted as the best and most thorough for BC tree species.
Purpose of the data	Required for calculation of baseline and project emissions

Comments	
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3.1.2 Data and Parameters Monitored

Data / Parameter	APRJ,i
Data unit	Ha
Description	Area of forest land in polygon, i
Source of data	See DARKWOODS_VRI_JUL2017_DistUpdate table in Darkwoods_VRI_Resultant_Jul2017 v1.1.mdb.
Description of measurement methods and procedures to be applied	Determined as part of the inventory work (see Darkwoods Inventory Update Report Darkwoods_VRIEM_Ecora2012.pdf)
Frequency of monitoring/recording	Annual
Value monitored	See DARKWOODS_VRI_JUL2017_DistUpdate table in Darkwoods_VRI_Resultant_Jul2017 v1.1.mdb.
Monitoring equipment	Visual, satellite, aerial photos
QA/QC procedures to be applied	GIS analysis is undertaken by experienced analysts utilizing standard and typical industry best practice, ArcGIS software, and local expertise and the latest NCC Standard Operating Procedures (SOP) – ‘Darkwoods Data Standards_28Aug17.doc’
Purpose of the data	Required for calculation of project emissions.
Calculation method	Area of polygons calculated using ArcGIS software
Comments	

Data / Parameter	APSP,i
Data unit	Ha
Description	Area of permanent sample plot in polygon, i
Source of data	Field measurement
Description of measurement methods and procedures to be applied	Standard variable radius plot layout design using field measuring tape from plot center. Methods outlined in ‘NCC Carbon PSP Plot Re-measurement SOP 2016 v1.0.pdf’ were followed for the re-measured plots and
Frequency of monitoring/recording	Plot measurements are repeated on 5-year intervals
Value monitored	See Plot Data in the submitted Excel file ‘Darkwoods plot data Jun 2017 v1.5.xlsx’

Monitoring equipment	GPS, measuring tape
QA/QC procedures to be applied	Latest NCC Standard Operating Procedures (SOP) followed, including check cruising processes. – ‘NCC Carbon PSP Plot Re-measurement SOP 2016 v1.0.pdf’ Specifically, this includes: 1) crew training to carefully follow the SOP document, 2) the establishment of blind check plots on 10% of the plots w/o access to original plot data, 3) the use of standardized equipment, and 4) keeping detailed data records on field sheets or data loggers.
Purpose of the data	Required for determination of Uncertainty Factor
Calculation method	Measuring tape. $Area = \pi \cdot r^2$
Comments	

Data / Parameter	DBH _{i,t}
Data unit	Cm
Description	Diameter at breast height measured for each tree in the sample plots at time, t
Source of data	Field measure
Description of measurement methods and procedures to be applied	Field measurements in permanent sample plots. Measurement with DBH tape for trees > 5 cm DBH.
Frequency of monitoring/recording	Individual plot tree re-measurements are repeated on 5-year intervals
Value monitored	See Plot Data in the attached excel file ‘Darkwoods plot data Jun 2017 v1.5.xlsx’
Monitoring equipment	DBH tape, data logger
QA/QC procedures to be applied	Latest NCC Standard Operating Procedures (SOP). – ‘NCC Carbon PSP Plot Re-measurement SOP 2016 v1.0.pdf’. Specifically, this includes: 1) crew training to carefully follow the SOP document, 2) the establishment of blind check plots on 10% of the plots w/o access to original plot data, 3) the use of standardized equipment, and 4) keeping detailed data records on field sheets or data loggers.
Purpose of the data	Required for determination of Uncertainty Factor and ultimately for calculation of project emissions.
Calculation method	Measured by hand
Comments	

Data / Parameter	Height _{i,t}
Data unit	M
Description	Tree height measured for each tree in the sample plots at time, t
Source of data	Field measurement
Description of measurement methods and procedures to be applied	Field measurements in permanent sample plots. Measurement with DBH tape for trees > 5 cm DBH. Used methods detailed in 'NCC Carbon PSP Plot Re-measurement SOP 2016 v1.0.pdf'
Frequency of monitoring/recording	Individual plot tree re-measurements are repeated on 5-year intervals
Value monitored	See Plot Data in the attached excel file 'Darkwoods plot data Jun 2017 v1.5.xlsx'
Monitoring equipment	Hypsometer, a transit, a clinometer, a relascope, a laser or other instrument designed for the measuring height.
QA/QC procedures to be applied	Latest NCC Standard Operating Procedures (SOP)– 'NCC Carbon PSP Plot Re-measurement SOP 2016 v1.0.pdf'. Specifically, this includes: 1) crew training to carefully follow the SOP document, 2) the establishment of blind check plots on 10% of the plots w/o access to original plot data, 3) the use of standardized equipment, and 4) keeping detailed data records on field sheets or data loggers.
Purpose of the data	Required for determination of Uncertainty Factor and ultimately for calculation of project emissions.
Calculation method	Measured using Vertex
Comments	

Data / Parameter	B _{AG i,t}
Data unit	t d.m. ha ⁻¹
Description	Aboveground live tree biomass in polygon, i, year, t, in the project case.
Source of data	Permanent sample plots
Description of measurement methods and procedures to be applied	Calculated from Height _{i,t} , DBH _{i,t} , and A _{PSPi}
Frequency of monitoring/recording	Upon establishment of PSP. Every 5 years, thereafter.

Value monitored	See Plot Data in the attached excel file 'Darkwoods plot data Jun 2017 v1.5.xlsx'
Monitoring equipment	See equipment for Height and DBH
QA/QC procedures to be applied	Latest NCC Standard Operating Procedures (SOP). 'NCC Carbon PSP Plot Re-measurement SOP 2016 v1.0.pdf'
Purpose of the data	Required for determination of Uncertainty Factor and ultimately for calculation of project emissions.
Calculation method	Above ground biomass for each tree within a permanent sample plot will be estimated from allometric equations using height and dbh (See 'Standish Biomass Equation' worksheet in 'Darkwoods plot data Jun 2017 v1.5.xlsx'). Area-based estimates of biomass are then be derived by summing all trees in plot and dividing by plot area.
Comments	Data used to validate ex-ante values from inventory + model output.

Data / Parameter	$B_{BG\ i,t}$
Data unit	t d.m. ha ⁻¹
Description	Belowground live tree biomass in polygon, i, year, t, in the project case.
Source of data	Derived from above ground biomass calculations within permanent sample plots.
Description of measurement methods and procedures to be applied	Calculated from $B_{AGi,t}$ and R_i
Frequency of monitoring/recording	Upon establishment of PSP. Every 5 years, thereafter.
Value monitored	See Plot Data in the attached excel file 'Darkwoods plot data Jun 2017 v1.5.xlsx'
Monitoring equipment	N/A
QA/QC procedures to be applied	Latest NCC Standard Operating Procedures (SOP).
Purpose of the data	Required for determination of Uncertainty Factor and ultimately for calculation of project emissions.
Calculation method	$B_{BGi,t} = B_{AGi,t} * R_i$
Comments	Data used to validate ex-ante values from inventory + model output.

Data / Parameter	$B_{TOTAL\ i,t}$
Data unit	$t\ d.m.\ ha^{-1}$
Description	Sum of $B_{AGi,t}$ and $B_{BGi,t}$
Source of data	Derived from above ground biomass calculations within permanent sample plots.
Description of measurement methods and procedures to be applied	Sum of $B_{AGi,t}$ and $B_{BGi,t}$
Frequency of monitoring/recording	Upon establishment of PSP. Every 5 years, thereafter.
Value monitored	See Plot Data in the attached excel file 'Darkwoods plot data Jun 2017 v1.5.xlsx'
Monitoring equipment	
QA/QC procedures to be applied	Latest NCC Standard Operating Procedures (SOP) – 'NCC Carbon PSP Plot Re-measurement SOP 2016 v1.0.pdf'
Purpose of the data	Required for determination of Uncertainty Factor and ultimately for calculation of project emissions.
Calculation method	Sum of $B_{AGi,t}$ and $B_{BGi,t}$
Comments	Data used to validate ex-ante values from inventory + model output.

Data / Parameter	$C_{LB\ i,t}$
Data unit	$t\ C\ ha^{-1}$
Description	Total carbon storage in live tree biomass in polygon, i, year, t, in the project case.
Source of data	Permanent sample plots.
Description of measurement methods and procedures to be applied	Calculated from $B_{TOTAL\ i,t}$ and CF
Frequency of monitoring/recording	Upon establishment of PSP. Every 5 years, thereafter.
Value monitored	See Plot Data in the attached excel file 'Darkwoods plot data Jun 2017 v1.5.xlsx'
Monitoring equipment	
QA/QC procedures to be applied	Latest NCC Standard Operating Procedures (SOP).
Purpose of the data	Required for determination of Uncertainty Factor and ultimately for calculation of project emissions.

Calculation method	$B_{TOTAL,i,t} * CF$
Comments	Data used to validate ex-ante values from inventory + model output.

Data / Parameter	$C_{DOM,i,t}$
Data unit	t C ha ⁻¹
Description	Total carbon storage in dead organic matter in polygon, i, year, t, in the project case.
Source of data	Permanent sample plots.
Description of measurement methods and procedures to be applied	Calculated from $DOM_{SNAG,i,t}$ and $DOM_{LDW,i,t}$ and CF
Frequency of monitoring/recording	Upon establishment of PSP. Every 5 years, thereafter.
Value monitored	See Plot Data in the attached excel file 'Darkwoods plot data Jun 2017 v1.5.xlsx'
Monitoring equipment	
QA/QC procedures to be applied	Latest NCC Standard Operating Procedures (SOP). – 'NCC Carbon PSP Plot Re-measurement SOP 2016 v1.0.pdf'
Purpose of the data	Required for determination of Uncertainty Factor and ultimately for calculation of project emissions.
Calculation method	$(DOM_{SNAG,i,t} + DOM_{LDW,i,t}) * CF$
Comments	Data used to validate ex-ante values from inventory + model output.

Data / Parameter	Mean tree age
Data unit	years
Description	Mean tree age for a given permanent sampling plot in polygon, i, for the project case.
Source of data	Permanent sampling plots
Description of measurement methods and procedures to be applied	Plot age is estimated by coring a sample of dominant trees during plot establishment
Frequency of monitoring/recording	Once during plot installation
Value monitored	See Plot Data in the attached excel file 'Darkwoods plot data Jun 2017 v1.5.xlsx'
Monitoring equipment	Tree coring bit

QA/QC procedures to be applied	Latest NCC Standard Operating Procedures (SOP)– ‘NCC Carbon PSP Plot Re-measurement SOP 2016 v1.0.pdf’. Specifically, this includes: 1) crew training to carefully follow the SOP document, 2) the establishment of blind check plots on 10% of the plots w/o access to original plot data, 3) the use of standardized equipment, and 4) keeping detailed data records on field sheets or data loggers.
Purpose of the data	Required for determination of Uncertainty Factor and ultimately for calculation of project emissions.
Calculation method	Counting visible tree rings
Comments	

Data / Parameter	$f_{PRJ,NATURAL,i,t}$
Data unit	unitless ($0 < f_{PRJ,NATURAL,i,t} < 1$)
Description	The proportion of biomass that dies from natural mortality in polygon, i, year, t, in the project case.
Source of data	Permanent sample plots, remote sensing
Description of measurement methods and procedures to be applied	Height and dbh of dead trees in permanent sample plots will be recorded. Areas of stand replacing fires.
Frequency of monitoring/recording	Annually in the case of fires, every 5 years in the case of individual plot trees
Value monitored	See Plot Data in the attached excel file ‘Darkwoods plot data Jun 2017 v1.5.xlsx’ in the case of individual plot trees; see Figure 2 in the case of fire boundaries
Monitoring equipment	Observation
QA/QC procedures to be applied	Latest NCC Standard Operating Procedures (SOP)
Purpose of the data	Required for project calculations
Calculation method	Tree mass components calculated from allometric equations (Standish, Manning, & Demaerschalk, 1985) and biomass expansion factors (Li, Kurz, Apps, & Beukema, 2003); (Lehtonen et al. 2004). Mass is converted to its carbon equivalent by multiplying by the carbon fraction (0.5). Proportion derived by comparison with calculated estimates of total carbon in polygon, i.
Comments	

Data / Parameter	$f_{PRJ,HARVEST,i,t}$
Data unit	unitless ($0 < f_{PRJ,HARVEST,i,t} < 1$)
Description	The proportion of biomass removed by harvesting from polygon, i, in year, t, in the project case.
Source of data	Darkwoods harvesting records
Description of measurement methods and procedures to be applied	harvest boundaries are mapped using field GPS methods and then overlaid with inventory data to determine how much biomass is removed from polygon i. New polygon boundaries may be formed when harvest boundaries don't match underlying inventory polygons.
Frequency of monitoring/recording	Annually
Value monitored	See harvest areas Figure 2
Monitoring equipment	GPS,
QA/QC procedures to be applied	Data will be verified by ground-truthing and comparison with remote sensing information.
Purpose of the data	Required for calculation of project emissions.
Calculation method	harvest boundaries are mapped using field GPS methods and then overlaid with inventory data to determine how much biomass is removed from polygon i. New polygon boundaries may be formed when harvest boundaries don't match underlying inventory polygons.
Comments	

Data / Parameter	$f_{PRJ,DAMAGE,i,t}$
Data unit	unitless ($0 < f_{PRJ,DAMAGE,i,t} < 1$)
Description	The proportion of additional biomass removed by for road and landing construction in polygon, i, year, t, in the project case.
Source of data	Remote sensing
Description of measurement methods and procedures to be applied	Areal estimate of removals derived from remote sensing data or ground based GPS measurements
Frequency of monitoring/recording	Annually
Value monitored	0
Monitoring equipment	Aerial photos
QA/QC procedures to be applied	Data will be verified by ground-truthing or remote sensing

	information.
Purpose of the data	Required for calculation of project emissions.
Calculation method	any new road polygons or landing are overlaid on inventory polygons and biomass that was removed from those areas is quantified.
Comments	

Data / Parameter	DOM _{SNAG,i,t}
Data unit	t d.m. ha ⁻¹ (d.m. = dry matter)
Description	Total mass of dead organic matter contained in standing dead wood in polygon, i, year, t in the project case.
Source of data	Permanent sample plots
Description of measurement methods and procedures to be applied	Calculated from Height _{i,t} , DBH _{i,t} , and A _{PSP,i} of dead trees measured in permanent sample plots
Frequency of monitoring/recording	Every 5 years
Value monitored	See Plot Data in the attached excel file 'Darkwoods plot data Jun 2017 v1.5.xlsx'
Monitoring equipment	na
QA/QC procedures to be applied	Latest NCC Standard Operating Procedures (SOP). – 'NCC Carbon PSP Plot Re-measurement SOP 2016 v1.0.pdf'
Purpose of the data	Required for determination of Uncertainty Factor and ultimately for calculation of project emissions.
Calculation method	Standing biomass for all snags within a permanent sample plot will be estimated from allometric equations using height and dbh (Standish, Manning, & Demaerschalk, 1985). (See 'Standish Biomass Equation' worksheet in 'Darkwoods plot data Jun 2017 v1.5.xlsx').
Comments	

Data / Parameter	DOM _{LDW,i,t}
Data unit	t d.m. ha ⁻¹ (d.m. = dry matter)
Description	Total mass of dead organic matter contained in lying dead wood in polygon, i, year, t in the project case.
Source of data	Permanent sample plots

Description of measurement methods and procedures to be applied	Calculated from using line intersect method (See – ‘NCC Carbon PSP Plot Re-measurement SOP 2016 v1.0.pdf’)
Frequency of monitoring/recording	Every 5 years
Value monitored	See Plot Data in the attached excel file ‘Darkwoods plot data Jun 2017 v1.5.xlsx’
Monitoring equipment	Tape and visual inspection
QA/QC procedures to be applied	Latest NCC Standard Operating Procedures (SOP). – ‘NCC Carbon PSP Plot Re-measurement SOP 2016 v1.0.pdf’
Purpose of the data	Required for determination of Uncertainty Factor and ultimately for calculation of project emissions.
Calculation method	Calculated using the following field- measured parameters $L_{i,t}$, $dn_{i,t}$, $D_{LDW,c,i,t}$, and $N_{i,t}$ as follows: $DOM_{LDW} = \sum M_{LDW,c}$ where, $M_{LDW,c}$ = the mass of LDW in density class, c ($t\ ha^{-1}$) $M_{LDW,c} = V_{LDW,c} * D_{LDW,c}$ where, $V_{LDW,c}$ = the volume per unit area calculated for each density class, c, $D_{LDW,c}$ = the density of LDW in density class, c ($t\ d.m.\ m^{-3}$)
Comments	

Data / Parameter	$V_{LDW,i,t}$
Data unit	$m^{-3}\ ha^{-1}$
Description	Total volume of dead organic matter contained in lying dead wood in polygon, i, year, t in the project case.
Source of data	Permanent sample plots
Description of measurement methods and procedures to be applied	Calculated from the line intersect method – ‘NCC Carbon PSP Plot Re-measurement SOP 2016 v1.0.pdf’
Frequency of monitoring/recording	Every 5 years
Value monitored	See Plot Data in the attached excel file ‘Darkwoods plot data Jun 2017 v1.5.xlsx’
Monitoring equipment	Tape and visual inspection

QA/QC procedures to be applied	Latest NCC Standard Operating Procedures (SOP). – ‘NCC Carbon PSP Plot Re-measurement SOP 2016 v1.0.pdf’
Purpose of the data	Required for determination of Uncertainty Factor and ultimately for calculation of project emissions.
Calculation method	Calculated using the following field-measured parameters $L_{i,t}$, $D_{n,i,t}$, and $N_{i,t}$ as follows: $V_{LDW,c} = \pi^2 * [(d_1^2 + d_2^2 \dots d_n^2)/8L]$ where, d_1, d_2, d_n = diameter (cm) of each of n pieces intersecting the line, and L = the length of the line (100 m default (Harmon, et al., 1986).d
Comments	

Data / Parameter	$L_{i,t}$
Data unit	m
Description	Calculation of lying dead wood: Length of the transect used to determine volume of lying dead wood in the sample plot, at time, t (default 100m)
Source of data	Permanent sample plots
Description of measurement methods and procedures to be applied	Field measurements
Frequency of monitoring/recording	Every 5 years
Value monitored	100m
Monitoring equipment	Tape
QA/QC procedures to be applied	Latest NCC Standard Operating Procedures (SOP). – ‘NCC Carbon PSP Plot Re-measurement SOP 2016 v1.0.pdf’
Purpose of the data	Required for determination of Uncertainty Factor and ultimately for calculation of project emissions.
Calculation method	Measuring tape
Comments	

Data / Parameter	$D_{n,i,t}$
Data unit	cm
Description	Calculation of lying dead wood: Diameter of each piece n

	of dead wood along the transects in the sample plot at time, t
Source of data	Permanent sample plots
Description of measurement methods and procedures to be applied	Lying dead wood must be sampled using the line intersect method (Harmon & Sexton, 1996). Two 50-m lines are established bisecting each plot and the diameters of the lying wood (> 10 cm diameter) intersecting the lines are measured. Minimum measurement diameter must not be less than 10cm.
Frequency of monitoring/recording	Every 5 years
Value monitored	See Plot Data in the attached excel file 'Darkwoods plot data Jun 2017 v1.5.xlsx'
Monitoring equipment	Caliper, diameter tape
QA/QC procedures to be applied	Latest NCC Standard Operating Procedures (SOP). – 'NCC Carbon PSP Plot Re-measurement SOP 2016 v1.0.pdf'
Purpose of the data	Required for determination of Uncertainty Factor and ultimately for calculation of project emissions.
Calculation method	Measuring tape
Comments	

Data / Parameter	$N_{i,t}$
Data unit	unitless
Description	Total number of wood pieces intersecting the transect in the sample plot, in time t.
Source of data	Field Measurement
Description of measurement methods and procedures to be applied	Lying dead wood is sampled using the line intersect method (Harmon & Sexton, 1996). Two 50-m lines are established bisecting each plot and the total number of wood pieces intersecting transect are counted.
Frequency of monitoring/recording	Every 5 years
Value monitored	See Plot Data in the attached excel file 'Darkwoods plot data Jun 2017 v1.5.xlsx'
Monitoring equipment	Visual observation
QA/QC procedures to be applied	Latest NCC Standard Operating Procedures (SOP). – 'NCC Carbon PSP Plot Re-measurement SOP 2016

	v1.0.pdf
Purpose of the data	Required for determination of Uncertainty Factor and ultimately for calculation of project emissions.
Calculation method	Field count
Comments	

Data / Parameter	D _{LDWc,i,t}
Data unit	t d.m. m ⁻³
Description	Basic wood density of dead wood in the density class, c along the transect in polygon, i, at time, t .
Source of data	Two 50-m lines are established bisecting each plot and wood pieces > 10 cm diameter intersecting transect are measured.
Description of measurement methods and procedures to be applied	Decay class is determined from visual and tactile inspection (See Darkwoods Project Description)
Frequency of monitoring/recording	Every 5 years
Value monitored	See 'Decay classes' worksheet in 'Darkwoods plot data Jun 2017 v1.5.xlsx'
Monitoring equipment	Visual and tactile inspection
QA/QC procedures to be applied	Latest NCC Standard Operating Procedures (SOP). – 'NCC Carbon PSP Plot Re-measurement SOP 2016 v1.0.pdf'
Purpose of the data	Required for determination of Uncertainty Factor and ultimately for calculation of project emissions.
Calculation method	Values for different decay classes derived from literature values from Heath and Chojnacky (1995) – softwood species
Comments	These have been compared to values reported for more local species in Harmon et al. (2011) and found to be consistent. Further a sensitivity analysis shows that the decay class wood densities can shift +/- 50% and still have no effect on the uncertainty factor.

Data / Parameter	E _M
Data unit	%
Description	An estimate of model error based on the relative area-weighted difference between of model-predicted values

	of carbon storage and those values measured in field plots
Source of data	Model output and field data
Description of measurement methods and procedures to be applied	Calculated value
Frequency of monitoring/recording	At each verification
Value monitored	-7.9%
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of the data	Required for determination of Uncertainty Factor and ultimately for calculation of project emissions.
Calculation method	Equation (60a) in Darkwoods Project Description Document. Calculation shown in the 'Uncertainty Factor' sheet in 'Darkwoods plot data Jun 2017 v1.5.xlsx'
Comments	

Data / Parameter	E_I
Data unit	%
Description	An estimate of Inventory sampling error calculated as the 90% confidence limit of the area-weighted differences between the model-predicted values of carbon storage and those values measured in field plots
Source of data	Model output and field data
Description of measurement methods and procedures to be applied	Calculated value
Frequency of monitoring/recording	At each verification
Value monitored	10.9%
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of the data	Required for determination of Uncertainty Factor and ultimately for calculation of project emissions.
Calculation method	Equation (60c) in Darkwoods Project Description Document
Comments	

Data / Parameter	E_P
Data unit	%
Description	An estimate of total project error based sum of the model and inventory error terms
Source of data	Model output and field data
Description of measurement methods and procedures to be applied	Calculated value
Frequency of monitoring/recording	At each verification
Value monitored	3.0%
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of the data	Required for determination of Uncertainty Factor and ultimately for calculation of project emissions.
Calculation method	Equation (60f) in Darkwoods Project Description Document
Comments	

Data / Parameter	$ER_{y,ERR}$
Data unit	%
Description	The uncertainty factor calculated for year 'y'
Source of data	Model output and field data
Description of measurement methods and procedures to be applied	Calculated value
Frequency of monitoring/recording	At each verification
Value monitored	1.5%
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of the data	Required for calculation of project emissions
Calculation method	As shown in Table 19 in the Darkwoods Project Description Document
Comments	

3.1.3 Monitoring Plan

Information regarding the organizational structure, responsibilities and competencies of the personnel that carried out the monitoring activities are provided in Section 2.4.1. Individuals responsible for specific activities are referenced in parenthesis using first and last initials (see Section 2.4.1 for reference). The organizational structure of 3GT is simple in that all members are directors responsible for the tasks outlined in Section 2.4.1. The organization structure for NCC is implicit in the position titles referenced in Section 2.4.1. Specifically, all staff report to Tom Swann and Rob Wilson who subsequently report to Michael Bradstreet. Information with respect to target precision levels, sample site locations, and stratification are shown in the subsection 4 of this section. With respect to the competencies of the key project team members, the 3GT team has extensive experience in managing VCS carbon projects and has been with this project since its inception. The NCC team also have extensive experience in establishing and maintaining large conservation projects. The specific degrees and titles achieved by each team member are reported in the team member table in Section 2.4.1.

The Darkwoods project monitoring plan contains three primary activities:

1. Annual inventory change monitoring (AL, NN, BS)

NCC has undertaken a non-systematic spatial monitoring on the entire property annually over the monitoring period, and the GIS inventory is updated annually. This was undertaken via a combination of aerial helicopter reconnaissance, field observations, field surveys of identified project activities, and field surveys of any identified disturbance events (i.e. fires, etc.). In addition, NCC has updated inventory databases with information from biodiversity and other non-carbon monitoring programs installed since 2008.

- a) Natural disturbance events (i.e. fires, high mortality pest and disease areas, blow-down areas, slides, etc.).

NCC has collected monitoring data on the spatial extent of annual forest fires during the monitoring period. There were a total of 5 measurable fires during the monitoring period, and only two > 2 ha in size (one in 2014, and one in 2015, See Table 2 and Figure 2). The fire areas were mapped using GPS while flying over in a helicopter. Output from the field GPS device was transferred to ArcGIS and incorporated into the Inventory database using standard overlay analysis. Data management was conducted according to the SOP document 'Darkwoods Data Standards_28Aug17.doc'. All fires, including several fire-starts that did not spread, were effectively managed by the provincial fire protection system. A map of the fire-starts and an overlay of the 2015 fire boundary on the 2016 satellite image from Google Earth are presented in Appendix 1. The 2015 overlay shows that the fire mapping was accurate.

Mountain pine beetle (MPB) activity had been in an epidemic condition in the property from 2008-2011. As such, new analysis units representing MPB-affected stands were developed and applied during the last verification period (2011-2012). Annual monitoring of insect-killed trees during the current monitoring period revealed that there is no longer any epidemic MPB activity within the Darkwoods area. There has been some minor individual tree mortality (also known as the endemic phase of MPB activity) in the area but this is included in stand-level modelling (See Darkwoods Project Description Document (v1.8) Appendix 1).

No material areas of blowdown, slides, or other (non-MPB) pests/disease were noted on the property.

Table 2. Darkwoods monitored fire events 2013-2016.

Project Year	# Fire Occurrences	Monitored Fire Occurrence (ha)
2013	0	0 ha
2014	4	80.8 ha
2015	1	102 ha
2016	0	0 ha

2. Planned project activities (i.e. harvests, road construction, reforestation, etc.) (AL, BS)

NCC undertook timber harvesting, as per the project scenario plan on an annual basis during the monitoring period. Harvest block are mapped either by walking the block with a handheld GPS device and then downloading the data to ArcGIS or by digitizing the block boundaries after the harvest using georeferenced aerial photos. Data management was conducted according to the SOP document 'Darkwoods Data Standards_28Aug17.doc'. The harvesting by monitoring period year is shown in Figure 2 and summarized in Table 3. The ex-post area data shown in Figure 2 are from digitized maps of actual Darkwoods harvest block areas. The projected volume was calculated by FORECAST/ATLAS assuming clearcut harvesting on the planned polygon area with the output. The ex-post delivered harvest volume is from BC Assessment Authority scaling data. There are some differences in the delivered vs. projected harvest volumes due to: 1) the fact that there is within block individual and grouped tree retention in the field that is not represented within the models; 2) small differences in the timing of reporting; 3) differences in projected and actual yields for a given polygon; and 4) differences in projected vs. actual merchantability. However, the simulation of clearcut harvesting with no tree retention is conservative from an ecosystem carbon storage perspective.

Figure 2. Harvesting and fire disturbances within the project area during the 2013-2016 monitoring period. The timber harvesting landbase is shown for reference and the two large fires are labeled.

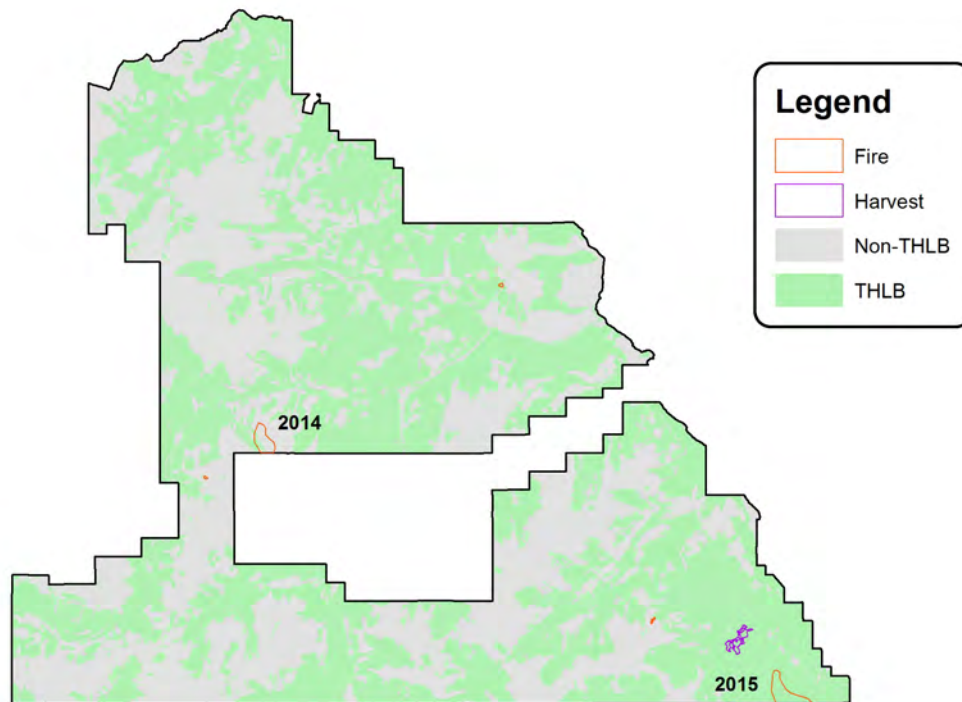


Table 3. Darkwoods Project Harvesting Activities 2013-2016

Project Year	Year	Ex-Post Projected Volume (m ³)	Ex-Post Delivered Volume (m ³)	Ex-Post Actual Area (ha)
6	2013	185	261	1.3
7	2014	0	0	0
8	2015	9,127	12,077	24.5
9	2016	5,138	5,486	13.8
	Total	14,450	17,825	39.6

3. Unplanned man-made disturbances (i.e. non-de minimis illegal or unplanned harvests, etc.): (AL,NN, BS)

No unplanned or illegal harvest or other activities were observed on the property during the monitoring period, thus no inventory updates or adjustments have been necessary. NCC has completed an access management plan, installed access controls on the property (i.e. gates), and implemented a mandatory permit system to monitor access on the property.

4. Other Monitoring Requirements of the Project

NCC will also document other monitoring requirements of the PDD, including:

a) Activity shifting leakage (annually) (RW, TS, MV)

NCC has not undertaken any material level of timber harvesting on any owned or managed property outside Darkwoods for the period 2013 to 2016, and therefore there is no risk of starting activity shifting risk (see Table 4). Annual records from all NCC properties are reviewed to determine if any harvesting activities have taken place. Records are kept by NCC (RW) for each property in an in-house online database. The results of this check are recorded in the monitoring report (Table 4).

Table 4. NCC Logging Activity on Other Properties (activity shifting risk) 2013-2016.

Property	Project Year	Logging Volume (m ³)	Activity shifting evidence/comment
All NCC properties outside Darkwoods	2013	0.0	n/a – no harvest
All NCC properties outside Darkwoods	2014	0.0	n/a – no harvest
All NCC properties outside Darkwoods	2015	0.0	n/a – no harvest
All NCC properties outside Darkwoods	2016	0.0	n/a – no harvest

b) Annual market leakage calculations (annually) (BS)

Annual Calculations of market leakage are shown for the years 2013-2016 in Section 3.2.3.

c) Field plot monitoring (RW, MV, AL, BS, TS, CW)

Field plot monitoring is described in detail below:

i) Sampling Design

Frequency of monitoring

Permanent plots are re-measured at intervals of not longer than 5 years (beginning at their date of first measurement). As noted, permanent plots may be established over multiple years, and such re-measurement schedules will be tracked on for each plot based on its establishment date.

Sampling conducted during this monitoring period

In fall of 2014 NCC completed the third phase of their field plot implementation program installing an additional 12 plots throughout the property to address some gaps identified during the last verification period. Specifically, the 2014 (phase 3) plot locations were selected based upon the updated inventory (2011) and with an objective of achieving representation across all analysis units. Sampling in the project landbase has been stratified such that each forest analysis unit has plot representation in proportion to its cumulative productive area in the project landbase (See Table 5). The coordinates of all sample plots are provided in the 'Plot Description' worksheet of the 'Darkwoods plot data Jun 2017 v.1.5.xlsx' spreadsheet. As stated in the PD, the number of plots established for the project area has been gradually increased since the validation such that the net change in carbon stocks should lie within 10 percent of the true value of the mean at the 90-percent confidence level. It should be noted that this 10% figure is intended as a guideline and not a hard threshold. The inventory and project error terms have been used to provide feedback for this estimate. The inventory error term sits at 10.9% and the project error term sits at 3.0% suggesting that the project is very close to the appropriate number of plots.

The field crews were given several sample plot locations using standard GIS software to generate the UTM coordinates of candidate polygons within the Darkwoods landbase. Crews followed the plot installation and measurement methods as outlined in the Standard Operating Procedures (SOP) document (Section 5.2). Plot size (radius = 4m, 10m, or 4m) was determined based upon the DBH range of the trees within the plots. Crews were instructed to establish a plot center and to measure the top height and DBH of each standing stem with a diameter ≥ 5 cm. Species and tree condition class (Live, Dead, Snag) were also recorded. 'Dead' indicated recently dead with branches still largely intact while 'Snag' indicated a standing dead stem with significant loss of branches. The crew also made note if a stem had a broken top and the recorded the height at break. Each plot also includes two 50M transects for the measurement of lying dead wood. While a total of 79 plots have been installed since the project start, 8 plots that were determined to be in the non-productive landbase, have been discontinued. Thus the project current has a total of 71 active plots. Further, the plot network is representative of the productive forest landbase within the Darkwoods project area and covers 19 of the 20 forest analysis units created for projecting forest carbon storage (Table 5). The only analysis unit without a plot is #110, which represents less than 0.01% of the productive landbase.

The plots installed during first and second phases of plot establishment (in 2011), were re-measured in 2016. During the re-measurements, the field crews attempted to locate all of the original tagged trees in the plots and to determine, standing and fallen dead trees, missing trees, and new trees (previously too small to be counted). The results of the field measurements are included in the attached excel file 'Darkwoods plot data Jun 2017 v1.5.xlsx'.

Table 5. Summary of the area of productive and non-productive forest by analysis unit.

C_AU	Productive Forest (ha)	Non-productive Forest (ha)	Actual plots	Expected plots ¹⁶
101	3,431.3	266.8	8	6.0
102	1,635.3	124.8	5	2.9
103	11,273.8	1,332.9	15	19.9
104	3,444.0	206.3	4	6.1
105	1,245.1	117.3	2	2.2
106	1,268.3	16.2	4	2.2
107	2,943.0	148.4	9	5.2
108	1,708.8	41.2	3	3.0
109	1,634.4	86.3	3	2.9
110	38.4	0.9	0	0.1
111		8,548.2	0	0.0
201	734.5	8.4	1	1.3
202	440.1	2.8	1	0.8
203	1,896.7	155.2	2	3.3
204	4,083.9	63.0	6	7.2
206	1,443.1	142.4	2	2.5
207	305.5	2.5	1	0.5
209	2,243.0	31.8	3	4.0
406a	166.8	26.7	1	0.3
406b	376.3	197.7	1	0.7
Totals	40,312	11,520	71	71

ii) Measurement and Data Analysis Techniques

Trees: Live & Standing Dead

Tree biomass per hectare for each plot was calculated from equations from (Standish, Manning, & Demaerschalk, 1985) that relate biomass to DBH and height for the 22 main tree species in British Columbia (as described in Section 3.4 of the PDD). The components for the Standish et al. (1985) equations were selected based upon tree class as shown in Table 6. A subset of 3 -5 dominant/co-dominant canopy trees within each plot were cored and aged to provide an estimate of stand age. The raw plot data for the each of the 71 plots are made available in attached excel file 'Darkwoods plot data Jun 2017 v1.5.xlsx'

Belowground biomass (t ha⁻¹) was calculated from live aboveground biomass using equations derived from (Li, Kurz, Apps, & Beukema, 2003) and (Lehtonen, Makipaa, Heikkinen, Sievanen, & Liski, 2004). Tree-level measurements (kg biomass per tree) were converted to area-based stand-level measurements (t ha⁻¹) and a conversion factor (0.50) was used to convert biomass into carbon. A detail description of the equations used in these calculations can be found in Section 3.4 of the PDD. The application of the

¹⁶ Expected number of plots is calculated by weighting the total number of installed plots based upon the total productive forest area in analysis unit relative to the total productive forest area. Eight plots that were determined to be in the non-productive forest landbase (C_AU 111) have been discontinued.

biomass and carbon calculations are shown in the available excel file 'Darkwoods plot data Jun 2017 v1.5.xlsx'.

Table 6. List of components used to calculate biomass in the Standish et al. (1985) biomass equations.

Class	Components used (a & b) ¹⁷	Belowground included ¹⁸
Live	DTOTAL	yes
Dead	DWOOD + DBRAN1	no
Snag	DWOOD	no

Lying Dead Wood

The mass of lying dead wood was measured in the permanent sampling plots using the line intersect method as described in the SOP document (Section 5.2). The diameters of all pieces of wood ≥ 5 cm diameter intersected by the transect lines were measured and recorded. Each piece of deadwood was assigned to one of three density classes, sound (0.32 t m^{-3}), intermediate (0.26 t m^{-3}), and rotten (0.15 t m^{-3}). The raw data and calculations are available in the file referenced in Section 5.2. A description of the detailed methodology and equations used in the calculation of lying dead wood is provided in the Section 3.3 of the PDD.

iii) Quality Assurance/Quality Control Measures (QA/QC)

The following QAQC standards were adhered to:

QA/QC for Field Measurements

NCC existing Standard Operating Procedures (SOP) with respect to field carbon measurement procedures were followed to ensure carbon field measurements were made accurately and reliably. Field crews were trained to be fully cognizant of all procedures and the importance of collecting accurate data.

At a minimum, 10% of the measured field plots will be check-cruised using blind checks with 100% re-measurement of all variables. Minimum thresholds in measurement error are as follows:

- 1) DBH (standing live and dead): +/- 10% standard error at 90% confidence interval
- 2) Height (standing live and dead): +/- 10% standard error at 90% confidence interval

¹⁷ The Standish et al. (1985) equations include different sets of component values depending on which biomass components are required. DTOTAL = total tree dry mass, DWOOD = stemwood dry mass, DBRAN1 = dry mass of large braches (>2.5cm)

¹⁸ Belowground biomass was estimated from aboveground biomass for live stems only. Dead stems and snags were assumed to have no live belowground biomass (only dead belowground biomass which is not included in the monitoring program).

3) Tree Count: +/- 10% standard error at 90% confidence interval

These are minimum thresholds for monitoring plot field accuracy, and will require re-measurement or re-establishment of plots as necessary to meet these requirements.

QA/QC for Data Entry

NCC standard procedures and ongoing QA/QC programs for data will be followed to ensure proper entry of data from paper to electronic format. If there are anomalies that cannot be resolved, the plot will be re-measured or omitted from the analysis.

Data gathered through the monitoring process were used for two main purposes:

- 1) to improve calculations of carbon stocks by fine-tuning the representation of each of the forest analysis units within the FORECAST model (see Section 3.4 of the PDD)
- 2) to evaluate the quality of the inventory and models through the calculation of the uncertainty factor described in Section 3.5 of the PDD

QA/QC for Data Archiving

NCC has document control procedures which adapted to cover the carbon monitoring data, including retaining the following for 2 years past the duration of the project:

- 1) Original copies of the field measurement, check plots, laboratory data, and related data summaries will be maintained in their original and electronic form
- 2) Copies of all monitoring data analyses, carbon calculations required for this methodology

QA/QC for Field Measurements conducted during the 2013-2016 monitoring period (AL, BS)

In the Fall of 2017 NCC undertook a blind check cruise program, which resulted in 7 plots being re-measured. DBH, Height, and Age measurements met the accuracy threshold of $\pm 10\%$ at 90% CI, as required by the PDD (See Section 5.2). The raw QAQC plot data and analysis are available in the file submitted Excel file 'QAQC 2017 Verification.xlsx'.

QA/QC for Data Entry for 2013-2016 monitoring period (AL, BS)

NCC standard procedures and ongoing QA/QC programs for data will be followed to ensure proper entry of data from paper to electronic format. If there are anomalies that cannot be resolved, the plot will be re-measured or omitted from the analysis (see SOP document 'Darkwoods Data Standards_28Aug17.doc').

QA/QC for Data Archiving for 2013-2016 monitoring period (AL, NN)

NCC has stored the electronic field data files, summary excel files, and age core samples at the Darkwoods field office (see SOP document 'Darkwoods Data Standards_28Aug17.doc').

3.1.4 Dissemination of Monitoring Plan and Results (CL3.2)

Information related to Darkwoods is disseminated via: <http://www.natureconservancy.ca/en/where-we-work/british-columbia/featured-projects/darkwoods/>.

NCC posts materials and results that are of interest to stakeholders for the project area on a periodic basis. See also Section 5.3.3 for additional information on communication of materials.

3.2 Quantification of GHG Emission Reductions and Removals

3.2.1 Baseline Emissions

Pre-stratification of Land Area

During the 2011-2012 project verification the inventory was updated and stratified into broad classes necessary to define the harvestable landbase according to regional standards. The following steps were taken in this process. These are similar to the steps that were taken with the previous inventory with a few minor exceptions as noted below.

1. Identify forested area - The identification of forested area was achieved by adding a new field 'Forested' and assigning polygons that contained tree species data in the field 'TREE_SPECIES_CODE_1' a value of "Y".
2. Identify operable landbase - The spatial boundary of the adjusted operable landbase (as defined in the PDD) was added to the new inventory using an overlay procedure.
3. Identify economically viable stands – Following regional standards and the approach outlined in the PDD, stands that would be considered uneconomical to harvest were identified. These included all hardwood stands and stands in which the leading species was either western hemlock, western red cedar, or subalpine fir and for which the stand age was >159y. In the 2008-2010 monitoring period, stands in age classes 8 or 9 (age >140) with the leading species mentioned above were assumed to be uneconomical. The 159y cutoff is more consistent with regional standards and feedback from Darkwoods foresters.
4. Identify road and stream buffers - Following regional standards and using existing spatial (linear) data for roads and streams within the Darkwoods property, road and stream buffers (polygons) were created to define area that was not forested or could not be harvested. Road buffers were created using a 5m buffer on either side of the centerline. Stream buffers were created in the same way using a 3m buffer.
5. Age adjustment for partially harvested stands – To avoid over estimating carbon storage, any stands identified as partially harvested in the past had their age reset to an age determined by the time of logging if the crown closure field was < 21%.

6. Identify non-productive forest area - Following regional standards forested polygons were classified either as productive or non-productive. The criteria outlined in Table 9 of the 2011-2012 Monitoring report were used to determine if stands were non-productive. In the previous inventory used in the 2008-2010 monitoring report, non-productive forest had already been identified in the spatial dataset. There was little documentation with respect to the criteria employed.

7. Correct for 2008-2010 harvesting in the baseline scenario – To avoid re-harvesting areas that had already been modeled as harvested in the baseline scenario modeled for the previous monitoring period (2008-2010), it was necessary to update the new inventory to reflect previous harvesting during this period. This was done by creating a copy of the new inventory dataset and updating it to account for the 2008-2010 spatial harvesting that was modeled for the baseline scenario in the previous monitoring report. Specifically, the spatial harvesting map created by FPS-ATLAS for the 2008-2010 period (based upon the old inventory) was overlaid upon the new inventory (see Figure 3 of 2011-2012 Monitoring report). Previously harvested polygons were identified in the new inventory and their ages were reset to 1 to indicate they had been harvested. In addition, analysis units were reassigned according to the transitions outlined in Table 11 of the 2011-2012 Monitoring Report. This corrected baseline inventory was used as the starting point for the projection of the current baseline scenario from 2010 onward. The original version of the new inventory (without the baseline harvesting corrections and which represents the ex-post project conditions) was used for the simulation of the project scenario. Both the baseline and project version of the new inventory are included in the spatial data set referenced in Section 5.2 of the 2011-2012 monitoring report.

The harvestable landbase was defined to include forested polygons that met all of the following criteria: 1) must be defined as productive forest, 2) must be defined as economical, and 3) must be within the boundary of the operable landbase. A spatial map of the harvesting landbase is shown in Figure 2.

Model Selection and Use

The FORECAST model (v8.4) and FPS-ATLAS model (v6.0.2.0) were used as the primary modelling tools for carbon stock changes calculations conducted at the stand and landscape scale. The combination of FORECAST and FPS-ATLAS meet all six criteria for model selection in the methodology document. In addition, these tools also meet preferred criteria #7 and #8. Further details about these models and their application in the Darkwoods Carbon Project are provided in the Darkwoods PD.

Calculating the Baseline Carbon Balance

The carbon accounting approach employed for the Darkwoods carbon project utilized the management interface and biomass output from a locally calibrated stand-level model, FORECAST, in conjunction with a forest-level harvest scheduling model, FPS-ATLAS.

FORECAST was used to simulate the temporal changes in carbon storage of different ecosystem pools for each the stand-level analysis units described below. The stand-level output from FORECAST was linked to the FPS-ATLAS model using a shared database library approach. This allowed the FPS-ATLAS landscape model to obtain information on the carbon pools associated with all of the inventory polygons that comprise the Darkwoods GIS database. FPS-ATLAS provided the framework and options for

simulating sequential harvesting on the landbase, as well as large-scale natural disturbance events for the baseline and project scenarios. In addition, it generated summary output of the carbon balance across the project area over time. Figure 8 in the Darkwoods PD provides an overview of the model linkages and their relationship to input sources and output data.

Stratification of Forested Area for Carbon Analysis

The project land base was stratified using an analysis unit approach as described in Section 4.1 of the PDD. Following the creation of the new inventory in 2011, the carbon analysis units were updated as described below.

The criteria used to assign the analysis units to the forested inventory polygons are shown in Table 7. A breakdown of the productive and non-productive forest area within each analysis unit is provided in Table 5. The regeneration assumptions used to simulate the analysis units in the FORECAST model are shown in Table 8; the transition rules following harvest are also provided. All of the analysis units were simulated using FORECAST v8.4 and effectively updated relative to the model output used in the 2008-2010 monitoring report. The specific carbon storage and yield curves generated for each analysis unit are included in the attached file 'Project Scenario Projection v1.1.xlsx'. For all analysis units, the conservative assumption was made that no further increment in volume or biomass would occur beyond a stand age of 250 years. Thus, when stands exceed that age in the ATLAS model, the values from year 250 are used.

Table 7. Assignment criteria for carbon analysis units.

AU	Description ¹⁹	Lead Species	Other Species	Prod Class	Site Index Range ²⁰	Age ²¹
none	non-forest					
101	Fd / L* / Bg/ Py_med	Fd / L* / Bg/ Py	PI	med	>12 <18	>40
102	Fd / L* / Bg/ Py_good	Fd / L* / Bg/ Py	H & C	good	>17	>40
103	B / L* / Fd / S* / Py / Pw_poor	B / L* / Fd / S* / Py / Pw	S, PI	poor	> 7 & <13, > 7 & <14 for S	>40
104	BL_med	BL	S, PI	med	>=13	>40
105	PL_poor	PL	B, S	poor	<13	>40
106	PL_med	PL	F, S	med	>12	>40
107	CW / HW_med-good	CW / HW	S, F	med-good	all	>40
108	S*_med	S*	BI, PI	med	>13 <17	>40
109	S*_good	S*	F, H, C	good	>16	>40
110	E / A_Med	E / A		Med	all	All
111	B / L* / Fd / S* / Py / Pw	B / L* / Fd / S* / Py / Pw			<8	All
201	F / L_med	F / L	PI	med	13-17	<=40
202	F / L_good	F / L	PI	good	>17	<=40
203	S/P/B_poor	S/P/B	L	poor	<13	<=40
204	S/B_Med	S/B	L	med	>12 for BI, >12 & <17 For S	<=40
206	P_med	P	F, S	med	>7	<=40
207	C / H_med-good	C/H	F, S	med-good	all	<=40
209	S_good	S	F, H, C	good	>16	<=40
888	Non-forest					
	MPB curves					
AU	Description	Lead	MPB Curve	Prod class	SI range	Age
406a	P_med	Fd/Hw/Py/Lw/PI	Y	med	>7	<86
406b	P_med	Fd/Hw/Py/Lw/PI	Y	med	>7	>85

¹⁹ F = Douglas-Fir, L = Western Larch, B = Subalpine Fir, S = Hybrid spruce or Engelmann Spruce, P = Lodgepole pine and White bark pine, C = Western red cedar, H = Western hemlock, E = Paper birch, and A = Trembling aspen.

²⁰ Site Index based on height at breast-height age 50.

²¹ The Age field is used to distinguish managed from natural stand types.

Table 8. Regeneration assumptions and transition rules (following harvest) for each analysis unit. Species composition, regeneration densities, regeneration delays (years following disturbance before regeneration occurs), and endemic mortality rates (percent of each species that are assumed to die from natural disturbance causes on 15-yr intervals). Transition pathways (to a managed stand AU) following simulated harvest by the ATLAS model are also shown.

Natural Stands													
AU	Description	SI	Sp 1	Sp 2	Sp 3	Sp1 %	Sp2 %	Sp3 %	Densit y	Regen delay	Endemic mort	Trans1 ²²	Trans2 ²³
101	F / L_med	16	Fd	Pl		75	25		1200	4	0.04	201	301
102	F / L_good	22	Fd	Hw	Cw	70	20	10	1600	3	0.04	202	302
103	B / L / F_poor	12	Bl	Se		75	25		1200	6	0.03, 0.05	203	303
104	B_med	16	Bl	Se		65	35		1800	4	0.03, 0.05	204	304
105	P_poor	12	Pl	Bl		70	30		1900	3	0.03	206	306
106	P_med	17	Pl	Se		70	30		2000	3	0.03	206	306
107	C / H_med-good	18	Hw	Cw		60	40		2000	3	0.06	207	307
108	S_med	16	Se	Bl	Pl	60	20	20	1500	6	0.03, 0.05	204	304
109	S_good	22	Sx	Fd	Cw	50	30	20	1600	5	0.04	209	309
110	E / A_Med	16	Ep	At		75	25		2000	2	0.03	110	110
111	B_very poor	7	Bl	Se		75	25		1000	6	0.03, 0.05	111	111
Managed Stands													
AU	Description	SI	Sp 1	Sp 2	Sp 3	Sp1 %	Sp2 %	Sp3 %	Densit y	Regen delay	Endemic mort	Trans1	Trans2
201	F / L_med	16	Fd	Pl		75	25		1500	1	0.04	201	301
202	F / L_good	22	Fd	Hw	Cw	70	20	10	2000	1	0.04	202	302
203	S/P/B poor	12	Se	Bl	Pl	40	35	25	1600	1	0.03, 0.05	203	303
204	S/P/B Med	16	Se	Bl	Pl	40	35	25	1600	1	0.03, 0.05	204	304
206	P_med	17	Pl	Se	Bl	60	30	10	2000	1	0.03, 0.05	206	306
207	C / H_med-good	18	Hw	Cw		60	40		2000	1	0.06	207	307
209	S_good	22	Sx	Fd	Cw	50	30	20	2000	1	0.04	209	309
Harvest Without Planting													
AU	Description	SI	Sp 1	Sp 2	Sp 3	Sp1 %	Sp2 %	Sp3 %	Densit y	Regen delay	Endemic mort	Trans1	Trans2
301	F / L_med	16	Fd	Pl		75	25		1000	6	0.04	301	301
302	F / L_good	22	Fd	Hw	Cw	70	20	10	1200	5	0.04	302	302
303	S/P/B poor	12	Se	Bl	Pl	40	35	25	800	8	0.03, 0.05	303	303
304	S/P/B Med	16	Se	Bl	Pl	40	35	25	1000	6	0.03, 0.05	304	304
306	P_Med	17	Pl	Se	Bl	60	30	10	1500	4	0.03, 0.05	306	306
307	C / H_med-good	18	Hw	Cw		60	40		1400	5	0.06	307	307
309	S_good	22	Sx	Fd	Cw	50	30	20	1000	7	0.04	309	309

²² Used in the simulation of the project scenario.

²³ Used in the simulation of the baseline scenario.

Calculation of annual emissions/reductions

Annual emissions/reductions for the baseline scenario for the years 2013-2016 were calculated using the output from the FPS-ATLAS model simulation (conducted as part of the 2011-2012 verification). The baseline scenario was modeled using the same assumptions laid out in Section 2.4 of the PDD. Generally speaking, the baseline scenario was based upon the assumption of 15-yr harvest depletion rate regime with a harvest target of 300,000m³/yr for 15 years followed by a short (1-2 year) gap in harvest activities before ongoing lower harvest levels resume. Output with respect to volume harvested and annual emissions/reductions (including key components) are shown in Table 9. Detailed FPS-ATLAS output is provided in the spreadsheet 'ATLAS Runs updated Dec 2012.xlsx'. Summarized output is provided in the spreadsheet 'Darkwoods Carbon Model 2013-2016 v1.2.xlsx'.

Table 9. Annual emissions/reductions for the baseline scenario calculated for 2013-2016.

Year	Ann. Baseline (Emissions) Reductions (tCO ₂ e)	Volume Harvested (m ³)	Baseline Net Change in Ecosystem C (tC)	Baseline Storage HWP (tC)	Baseline Storage Waste Products (tC)	Baseline (Emissions) Production & Waste (tC)
2013	(241,227)	301,285	(67,012)	3,843	1,238	(3,858)
2014	(285,826)	300,673	(79,173)	3,835	1,235	(3,850)
2015	(316,975)	301,390	(87,671)	3,844	1,238	(3,859)
2016	(257,335)	307,123	(71,429)	3,917	1,262	(3,932)

3.2.2 Project Emissions

Stratification of Land Area

The project land base was stratified using the same analysis unit approach as described in Section 3.2.1.

Calculation of annual emissions/reductions

Annual emissions/reductions for the project scenario for the years 2013-2016 were calculated using a spreadsheet-based model to simulate the aging of the forested polygons in the Darkwoods inventory (Darkwoods_VRI_Resultant_Jul2017v1.1.mdb) forward from 2011 to 2016. The harvest and fire events that occurred in the project landbase during the 2013-2016 period (See Fig. 2) were represented in the simulation using a three-step process:

1) The spatial disturbance data (see attached shape file) were intersected with the previous spatial inventory data (Darkwoods_VRI_Resultant_Oct2012_v2.3.mdb) using ArcMap v10.1 to create an

updated inventory file (Darkwoods_VRI_Resultant_Jul20171.1.mdb). The new file includes the identification of current disturbed forest area (harvest and fire).

2) New annual fields were added to the inventory to represent the condition (C_AU_yrX, Age_yrX, Net ecosystem C content_yrX, and Net merchantable volume_yrX) of each forested polygon for each year starting at 2011 and going through 2016.

3) The new fields were updated (including C_AU transitions, age increments, and carbon and volume contents) based upon the timing of the disturbance events and the transition patterns shown in Table 10. The transition patterns show which C_AU a specific polygon will transition to following a harvest or fire event. These patterns were derived using a 2-step process. In the first step, the age of the stands within the burned polygons were evaluated to estimate the amount tree biomass present at the time of the fire. It should be noted that all natural stand analysis units included in the analysis are assumed to have originated following a stand replacing fire (see Section 4.1 of the Darkwoods PD). This is important as it influences the amount of dead biomass left behind following the fire. In the second step, the determination is made as to whether a new natural stand analysis unit (with adjusted dead wood biomass pools) needs to be created. If the age of the burned was reasonably close to the age of the trees in the precursor stand used to simulate the natural stand analysis unit associated with that polygon, the original natural stand analysis unit is retained and the age of the 'new' stand in the next year following a disturbance event will be reset to 0. In contrast, if it was determined that a stand was much younger or older than that represented in the precursor of the natural stands, a new C_AU is assigned with appropriately adjusted the dead wood pools that reflect the age at burning. The age analysis and the new curves are shown in 'Fire summary' and 'Fire Curves' worksheets of the 'Project Scenario Projection v1.1.xlsx' document, respectively.

The spreadsheet-based model and all associated stand curves, transition patterns, calculations, etc. are contained in the attached file 'Project Scenario Projection v1.1.xlsx'. As in the baseline modelling, for all analysis units, the conservative assumption was made that no further increment in volume or biomass would occur beyond a stand age of 250 years. Thus, when stands exceed that age in the spreadsheet model, the values from year 250 are used. The starting carbon contents calculated for each C_AU included in the inventory (see Tables 5, 8, and 9) were compared against the values generated by the ATLAS Model for the 2011-2012 verification to ensure consistency (Table 11). During this process we discovered a small error in the assignment of C_AUs in the ATLAS project 2012 database used in the previous analysis (2011-2012 verification). Specifically, polygons that should have been assigned to the 406a and 406b forest units were mistakenly assigned to the non-forested 888 unit. This mistake was not made in the ATLAS baseline database from 2012. The error led to an underestimate of project carbon during that period as thus was conservative. The error has been corrected in the updated inventory used in the Project Scenario Simulator spreadsheet model. Otherwise the two datasets were consistent.

Once the dataset was verified, the Project Scenario Simulator was used to calculate changes in net ecosystem carbon content for the years 2013-2016 in the project scenario. Prior to completing the simulation it was necessary to create a new set of fire curves to represent conditions after the fires that occurred in during 2014 and 2015. These are included in the file 'Project Scenario Projection v1.1.xlsx' in a worksheet entitled 'Fire Curves'. Each of the fields in the spreadsheet model include comments describing the data they contain and their use in the calculations. Of note, the timing of disturbances (fire and logging) that occurred in 2011-2012 are included in the "Project_Harvest_Year" field. Alternatively,

the "Year_" field was used to identify the timing of fire or harvest events that occurred during the current verification period (2013-2016). In addition, the "Disturb_Ty" field indicates the type of disturbance that occurred between 2013 and 2016, the "C_AU" field indicates the analysis unit as of the end of 2010, and the "C_AU_post2011_Log" field indicates the analysis unit after the 2011 logging season. Finally, the "C_Age_2011" field indicates the age of a specific polygon as of the end of 2011.

The spreadsheet model determines the net ecosystem carbon content for each forested polygon (i) and each time period (Px) in the Darkwoods landbase (represented in the rows of the 'Darkwoods_VRI_Jul2017 v1.1' worksheet) using the C_AU_Px fields in combination with simulated stand ages for each time period (shown in the C_Age_Px fields) to look up values of carbon storage (t C ha⁻¹) from specific stand curves (analysis units) included in the 'Ecosystem C' worksheet. The net ecosystem carbon content for each polygon (i) and time period (Px) is then calculated by multiplying the area of the specific polygon (i) by the carbon storage per ha. These values are reported in the fields entitled NetEcoC_Px. Finally, total net carbon storage is summarized by analysis unit and simulation year using a pivot table function linked to the data in the 'Darkwoods_VRI_Jul2017 v1.1' worksheet and reported in the 'Summary' worksheet.

The calculated values for net ecosystem carbon content and harvested volume are shown in Table 12. As described in Section 2.2.2, provincial scaling reports were used to quantify the amount of wood volume harvested from the Darkwoods landbase during each year of the verification period as these data should be more accurate than model projections. Volume data were converted to biomass assuming an average wood density of 0.4 t m⁻³. Specific calculations are shown in Table 4 in the 'Summary Tables & Figures' worksheet of the 'Darkwoods Carbon Model 2013-16 v1.2.xlsx' spreadsheet provided. Calculations for carbon storage in harvested wood products, waste products and GHG emissions related to equipment and production are shown in Table 5 in the 'Summary Tables & Figures' worksheet. These values were determined using the equations and associated parameter values shown in the 'HWP Carbon Model' worksheet of the 'Darkwoods Carbon Model 2013-16 v1.2.xlsx' spreadsheet. The justification for the equations and parameters is provided in Sections 8.1.6 through 8.1.9 of the VM0012 Methodology.

Table 10. Transition of forested polygons following disturbance events in the Project Scenario Simulator spreadsheet model.

Initial	After Harvest	After Fire ²⁴
C_AU	C_AU	C_AU
101	201	1101
102	202	1102
103	203	1103
104	204	1104
105	206	1105
106	206	1106
107	207	1107

²⁴ The attributes curves for the fire C_AUS are included in the Fire Curves worksheet of the 'Project Scenario Projection v1.1.xlsx' file.

108	204	1108
109	209	1109
110	210	1110
111	203	1111
201	201	1201
202	202	1202
203	203	1203
204	204	1204
206	206	1206
207	207	1207
209	209	1209
406a	206	1406
406b	206	1406
888	888	888

Table 11. Comparison between 2011 carbon contents by C_AU between the ATLAS Project Database used for the 2011-2012 verification to the Project Scenario Simulator used for the 2013-2016 verification. The analysis units showing differences are highlighted.

	ATLAS Project Database		Project Scenario Simulator	
C_AU	ha	NetEcoC (t C)	ha	NetEcoC (t C)
101	3,698.1	552,774	3,698.1	552774
102	1,760.1	342,879	1,760.1	342879
103	12,606.7	1,386,272	12,606.7	1386272
104	3,650.3	498,131	3,650.3	498131
105	1,362.4	116,800	1,362.4	116800
106	1,284.5	192,124	1,284.5	192124
107	3,091.5	689,228	3,091.5	689228
108	1,750.0	354,829	1,750.0	354829
109	1,720.7	318,712	1,720.7	318712
110	39.3	4,154	39.3	4154
111	8,548.2	663,073	8,548.2	663073
201	742.9	31,515	742.9	31515
202	442.9	23,014	442.9	23014
203	2,051.9	52,990	2,051.9	52990
204	4,146.9	129,744	4,146.9	129744
206	1,585.5	70,664	1,585.5	70664
207	308.0	14,733	308.0	14733
209	2,274.8	96,768	2,274.8	96768
406a	0.0	0	193.5	32309
406b	0.0	0	574.1	62572

888	3,764.3	0	2,996.8	0
Total	54,829.0	5,538,406	54,829.0	5,633,287

Table 12. Annual emissions/reductions for the project scenario calculated for 2013-2016.

Year	Ann. Project (Emissions) Reductions (tCO ₂ e)	Volume Harvested (m ³)	Project Net Change in Ecosystem C (tC)	Project Storage HWP (tC)	Project Storage Waste Products (tC)	Project (Emissions) Production & Waste (tC)
2013	130,376	261	35,556	3	1	(3)
2014	119,893	0	32,698	0	0	0
2015	50,274	12,077	13,662	154	50	(155)
2016	157,455	5,486	42,920	70	23	(70)

3.2.3 Leakage

Market Leakage

The market leakage factors were calculated using Option 2 – CAR Leakage Calculation (see Darkwoods PD for a complete for description). In general, the calculations for the Option 2 Leakage were conducted using output for annual timber harvest volume data from the Atlas/Forecast modeling for the baseline scenario (see Table 9) and from the regional annual harvest scaling reports for the project scenario (see Table 12 and related text in Sections 2.2.2 and 3.2.2). The leakage for each year (LEy) was calculated as the annual net change in harvest volume between the baseline and project scenario, in tCO₂e, multiplied by 20% (as per Figure 2 in the VM0012 Methodology and the related calculations - shown in Figure 13 of the Darkwoods PD). The specific calculations used to derive the values shown in Table 13 can be seen in Table 7 in the 'Summary Tables & Figures' worksheet in the 'Darkwoods Carbon Model 2013-2016 v1.2.xlsx' spreadsheet included.

This market leakage option does not require further monitoring activities, other than monitoring actual scenario harvest volumes (shown in Table 12). The resulting leakage risk factors determined for each monitoring period year are shown in Table 13.

Table 13. Annual Market Leakage Discounts for the 2013-2016 monitoring period.

Year	Change in carbon associated with harvested wood (tCO ₂ e)	Leakage Risk Discount (tCO ₂ e)	Leakage Risk % of VCUs
2013	220,751	(44,150)	11.9%
2014	220,494	(44,099)	10.9%

2015	212,163	(42,433)	11.6%
2016	221,200	(44,240)	10.7%

3.2.4 Net GHG Emission Reductions and Removals

The determination of net GHG Emissions and the associated calculation of total saleable VCU were made following the approach and equations described in Section 4.4 of the VCS PDD. The net change in ecosystem carbon stocks (Table 14a) was calculated by subtracting the annual changes in net ecosystem C in the project scenario from those in the baseline scenario. The specific calculations made to derive these values can be found in Table 3 of the 'Summary Tables & Figures' worksheet in the 'Darkwoods Carbon Model 2013-2016 v1.2.xlsx' spreadsheet. These values were subsequently utilized to determine the net emissions reductions and removals shown for the 2013-2016 verification period in Table 14a and Table 15. The calculations made to determine these values can be found in the spreadsheet 'Darkwoods Carbon Model 2013-2016 v1.2.xlsx'.

Table 14a. Net changes in ecosystem C content calculated by subtracting the baseline scenario from the project scenario.

Year	Baseline TOTAL Ecosystem C (tC)	Baseline CHANGE Ecosystem C (tC)	Project TOTAL Ecosystem C (tC)	Project CHANGE Ecosystem C (tC)	Net CHANGE Ecosystem C (tC)
2013	5,112,997	(67,012)	5,737,692	35,556	102,568
2014	5,033,824	(79,173)	5,770,390	32,698	111,871
2015	4,946,153	(87,671)	5,784,052	13,662	101,333
2016	4,874,724	(71,429)	5,826,972	42,920	114,349
Total		-305,285		124,836	430,121

Table 14b. Summary of emission removals and leakage emissions for the baseline and project scenarios for the 2013-2016 verification period. The adjusted net emission reductions column represents the emission reductions associated with net change in carbon stocks between the baseline and project scenarios.

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Gross GHG emission reductions or removals (tCO ₂ e)	Adjusted Net GHG emission reductions or removals (tCO ₂ e)
2013	(241,227)	130,376	371,602	357,470
2014	(285,826)	119,893	405,719	391,603
2015	(316,975)	50,274	367,248	353,666
2016	(257,335)	157,455	414,790	400,629
Total	(1,101,363)	457,997	1,559,360	1,503,369

Table 15. Total saleable VCUs and terms used in their calculation for the 2013-2016 verification period.

Year	Ann. Gross Emissions Reductions (tCO ₂ e)	Leakage Risk Discount (tCO ₂ e)	Uncertainty Risk Discount (tCO ₂ e)	Annual Buffer Set-Aside (tCO ₂ e)	Cumulative Buffer Account (tCO ₂ e)	Release From Buffer (tCO ₂ e)	Saleable VCU (tCO ₂ e)
2013	371,603	(44,150)	(4,912)	(35,747)	227,288		286,793
2014	405,719	(44,099)	(5,424)	(39,160)	266,448		317,035
2015	367,248	(42,433)	(4,872)	(35,367)	301,814		284,577
2016	414,790	(44,240)	(5,558)	(40,063)	290,596	51,282	376,211
Total	1,559,360	(174,922)	(20,767)	(150,337)	290,596	51,282	1,264,617

3.2.5 Calculation of the Uncertainty Factor

As described in Section 3.5 of the PDD, an Uncertainty Factor including inventory and model error terms were calculated based upon the comparison of the stand-level measures of net aboveground ecosystem C within the field plots and the associated model-projected values. A total project error (EP) of 4.0% was determined as the sum of the calculated Inventory error (EI) term of 10.9% and a model error (EM) term of -6.9%. Thus, the value of the Uncertainty Factor was determined to be 1.5% based upon equation below. The model error term was slightly negative (under predicting carbon storage) relative to the inventory plots.

As described in Section 3.5 of the PDD the Uncertainty Factor is determined from the inventory and model error terms as follows:

Uncertainty Factor =

If EP < 10% then 1.5%;

If EP > 10% then 1.5% + EP – 10%

where:

EP = EI + EM

EP = Total project error (%)

EI = Inventory error at the 90% confidence interval (% , expressed as the absolute value of the ± error term).

EM = Model error (%). May be positive (model > measured value) or negative (model < measured value).

The spreadsheet used in the calculation of the Uncertainty Factor is attached 'Darkwoods plot data Jun 2017v1.4.xlsx'.

3.3 Optional Criterion: Climate Change Adaptation Benefits

Not Applicable.

3.3.1 Activities and/or Processes Implemented for Adaptation (GL1.4)

Not Applicable.

4 COMMUNITY

4.1 Net Positive Community Impacts

4.1.1 Community Impacts (CM1.1)

The Darkwoods project area is the private, fee simple property with gated road access control. There are no communities, indigenous people or other public communities or ongoing public use for income, livelihood, or critical cultural values by any people living or dependent on resources within the project area that would be affected by project activities during this monitoring period.

1. Community project area use:

There are 29 non-permanent recreational cabins/trailers located on a small area of the property known as Tye. This usage is declining (from 35 in 2013) as leases expire and are not renewed, but these leases were not affected by project activities during the monitoring period.

There are a few (2-4) small (3-4 worker) exploration and mining/prospecting operators occasionally active on the property. The project implementation did not affect these users during the monitoring period.

More generally, a limited set of people from the local communities interact directly with the Darkwoods property area on an intermittent basis via:

Seasonal or project-based contracting opportunities for local road construction, maintenance, and logging contractors,

Project based forest management consulting for inventory, planning, monitoring, or other project related work

Backcountry recreational use

Researchers undertaking scientific studies

These users fluctuate within the monitoring period years based on project activity levels, projects, and general community usage. Overall there has been no specific changes to project activities or policies that has impacted these community users during this monitoring period.

The estimated net impacts of actual project activities in the monitoring period are outlined in Section 4.3 below.

2. Project Zone Community:

The CCB PDD identified the Kootenay Lake Timber Supply Area as a relevant spatial extent for the project zone, and no changes are proposed to this designation in this monitoring report. An estimate of the net impacts of actual project activities are outlined in Sections 4.2 and 4.3 below.

3. Direct employment on Darkwoods:

During the monitoring period NCC continued to employ 1 full time site supervisor, along with a fairly constant level of regional and national staffing related to Darkwoods. There has been no material change to the direct employment benefits related to Darkwoods during this monitoring period.

4. Community Values

The CCB PDD demonstrated that the local communities highly value the environment, the regions natural beauty and amenities, and the related quality of life. There is evidence that the quality of the local environment attracts people and businesses to the area, drives a high level of economic diversity into the area, and may positively benefit real estate values and other positive local community features. The Darkwoods project is a significant area that contributes directly to this amenity value for the local communities. The project has continued to provide this value to the community during this monitoring period, and although no new specific community survey data is available, it appears logical that there has been no material change to these community attitudes and values during the monitoring period.

5. First Nations:

There has been no change to NCC's ongoing relationship with the Ktunaxa Nation during this monitoring period. There is no material ongoing use or special cultural values on the project area that are impacted by project activities.

4.1.2 Net Positive Community Well-Being Impacts (CM1.1)

As there are no people or communities within the project area, there are no direct community impacts in either the with- or without-project scenarios on the project area.

As described in the CCB PDD, the without-project scenario would have generated economic, climate services, other ecosystem services, and amenity value impacts that would have both positively and negatively impacts communities in the surrounding project zone.

Implementation of the baseline scenario would have:

Generated economic benefits for select community members due to the increased labour and income related to higher harvesting activity levels;

Generated negative impacts by foregoing the investment and diversified economic activities in the project scenario, negatively impacting climate services, negatively impacting other ecosystem services (including water), and negatively impacting regional amenity values that are important to community members and community development.

As noted in the CBB PDD, Section G1 and G2.4, the project zone is a highly diversified modern economy that has a low dependence on forestry. Additionally, the forest industry makes up a relatively small component of the project zone economies, small component of project zone employment, and is a declining source of employment over the past several decades. During this monitoring period, the higher harvest levels in the baseline scenario would likely have expanded local forestry economic opportunities, but a substantial portion of these benefits would have accrued to stakeholders outside the project zone as

a portion of services are provided by mobile outside contractors and a substantial volume of the harvested wood would have been delivered to external facilities.

In addition, the CBB PDD discusses the fluid and flexible nature of regional log markets and regional log flows, and posits that on average the market would adjust to the presence or absence of the baseline harvest volume without spiking or growing employment in a direct linear manner. In reality, the regional log markets are very interconnected, and are more reactive to national and international market conditions than by small fluctuations in local log supply. We expect in reality the higher harvest levels in the baseline scenario would result in the market deferring higher cost and/or lower quality wood sources in the regional fiber basket (particularly on long term crown tenures), or otherwise changing log flows to be more cost optimal with the Darkwoods volume in the mix rather than just expanding to the supply. In other words, the potential benefits to community stakeholders in direct employment and economic opportunity would likely not expand in direct relationship to higher volume, but rather the market would substitute Darkwoods volume for other less optimal volume while keep production levels relatively similar. That said, to be conservative (and due to lack of direct supporting evidence) for this impact analysis we have assumed the entire Darkwoods baseline harvest volume contributes additive employment and economic opportunity to the project zone.

Comparatively, the implementation of the project has:

Contributed direct economic development through project activities, but negatively impacted the economic benefits of the higher timber harvesting in the baseline scenario (due to lower harvest levels in the project);

Generated substantial climate services benefits;

Generated substantial other ecosystem services benefits, including water provisioning and quality, valuable fisheries contributions, potentially pollinator support, and other ecosystem services;

Enhanced the community amenity values for the local region, including visual qualities, recreational opportunities, scientific study, and supporting regional environmental values and issues.

Key Results:

There is no impact on communities on the project area (no communities present).

Based on the monitoring results shown in Section 4.3, and as shown in Table 16, project implementation has had a net positive impact on community well-being during this monitoring period.

The baseline scenario generates positive direct economic employment opportunity for the community during this monitoring period, but this is offset by the negative climate services value impact, along with qualitative negative impacts on other ecosystem services and community amenity values.

The project scenario generates positive direct economic employment opportunities (although significantly less than the baseline), but also generates significant climate services value, along with qualitative additional benefits from other ecosystem services provisioning and community amenity benefits.

In total, implementing the project provides a total net benefit to the communities, primarily through the climate services benefits outweighing the higher direct economic opportunities in the baseline. The baseline generates a net economic gain of \$5.75 million and -6 other values versus the project generates a net economic gain of \$14.41 million and +10 other values.

The ecosystem services provisioning and community amenity values are assessed qualitatively because of cost and complexity to reliably estimate a value for this on a project level. However, although we used a simple +/- benefit rating score, that does not mean these value are not highly valuable if properly valued. In other words, the reader should not discount the potentially high economic value of these qualitative values, which the project scenario clearly provides a net benefit over the baseline scenarios.

Finally, not that these results are representative of this monitoring period, where the baseline scenario is at peak harvesting production, and the project scenario comparatively below expected harvest levels. Over the project lifespan, the economic benefits from the baseline will sharply decline (after year 15) and become very comparable to the project scenario in direct economic benefits in the long run.

Table 16. Net community benefits assessment by scenario

Project Area Net Impacts:

Baseline Scenario		Project Scenario	
# People	Impacts	# People	Impacts
0	None	0	None

Project Zone Net Impacts (total monitoring period, 2013-2016)

	Baseline Scenario				Project Scenario			
	Activity (Person Years)	Economic Impact (\$CDN)	# Positive Impacts	# Negative Impacts	Activity (Person Years)	Economic Impact (\$CDN)	# Positive Impacts	# Negative Impacts
Employment	809.8 ²⁵				57.0 ²⁶			
Economic Benefit		\$34.52				\$2.43		
Climate Services Value		\$(28.77)				\$11.98		
Sub-Total Economic Value:		\$5.75				\$14.41		
Ecosystem Services (net count)			1	-6			9	0
Amenity Services			0	-1			1	0
Net Services Impact Total:				-6 (-)				10 (+)

4.1.3 Protection of High Conservation Values (CM1.2)

Not applicable.

4.2 Offsite Stakeholder Impacts

4.2.1 Mitigation of Negative Impacts on Other Stakeholders (CM2.2)

No additional specific steps were necessary or taken to further mitigate negative impacts on outside stakeholders, as the expected net benefits will be positive; similar to the project zone assessment summarized in Section 4.1.

4.2.2 Net Impacts on Other Stakeholders (CM2.3)

As described in the CCB PDD, the impacts to offset stakeholders can be reasonably assumed to be similar (or proportional) to the impacts to the project zone stakeholders. CCB PDD data estimated 50.4%

²⁵ See Table 18.

²⁶ Project PY calculated by adding results shown in Table 18 and Table 19.

of the forestry and timber processing in the project zone (Kootenay Lake TSA) accrue to stakeholders outside the project zone, and climate services, ecosystem services provisioning, and some portion of the community amenity values will also benefit stakeholders outside the project zone.

Therefore, the net benefits described in Section 4.1.2 and the results in 4.3.2 act as proxies for the net impact on other stakeholders outside the project zone as well. Therefore, the project implementation will be a net benefit in a manner similar to the project zone stakeholders.

There is no change to this assessment of the similarities of impacts on other stakeholders during this monitoring period and no additional unique impacts were identified during this period.

4.3 Community Impact Monitoring

Community impact monitoring is an analysis that is undertaken at each verification, and calculated and reported on an annualized basis. This analysis is not undertaken on a specific date, but completed as needed for a monitoring report. For this monitoring report, the analysis was undertaken in the period of July-August 2017.

4.3.1 Community Monitoring Plan Development (CM3.3)

Not applicable.

4.3.2 Community Monitoring Plan Results (CM3.1, CM3.2, GL2.5)

1. Community Impact Monitoring Activity 1: Economic/employment impact

- The monitoring for economic and employment impacts is done by re-calculating an estimate of labour and employment income generated from annual baseline and project scenario harvesting activities for the years in this monitoring report.
- The same factors are applied as in the CBB PDD as shown in Table 17.

Table 17 - Key Harvesting Employment Variables Within the Project Zone^{27, 28}

Average Harvested Volume (Kootenay Lake TSA; all ownerships)	754,953 m ³ /year
Estimated Direct Harvesting, Planning, and Silviculture Employment (Kootenay Lake TSA)	258 PY (Person Years)
Estimated Direct Primary Processing Employment (Region wide; generated by Kootenay Lake harvest volume)	494 PY (Est. 249 PY within Kootenay Lake TSA)
Total Estimated Direct Forest Industry Employment	846 PY
Employment Coefficients:	
Harvesting/Silviculture (employment generated in Kootenay Lake TSA)	0.342 PY/1000m ³
Primary Processing (employment generated in Kootenay Lake TSA) ²⁹	0.327 PY/1000m ³
Total employment generated from Kootenay Lake TSA harvest volume	0.669 PY/1000m ³
Average employment income from forestry and wood processing in the Kootenay Lake TSA (\$2005)	\$42,630/year

Summarized, during this monitoring period:

The baseline scenario would have generated economic benefits from the significantly higher harvest levels in the initial 15 years. This harvest volume is estimated to generate 202.5 person years of employment per year, on average, or 809.8 person years total for the monitoring period. This employment would generate an estimated \$8.6 million per year in employment income, or \$34 million over the monitoring period.

The project scenario harvested significantly less volume, but also additional project scenario economic benefit. In total (see Table 18 and Table 19), the project is estimated to generate direct employment of 14.3 person years per year, on average; and 57.0 person years total for the monitoring period. This employment is estimated to generate \$0.61 million per year, and \$2.43 million over the monitoring period.

In total, and as expected, the high level of initial harvest in the baseline scenario would likely have generated substantially more economic opportunity for the project zone during this monitoring period. As

²⁷ SIBAC. 2009. Southern Interior Beetle Action Coalition Kootenay Lake TSA Socio-Economic Trend Assessment. Merritt, BC.

²⁸ Lefebvre, P. 2008. Central Kootenay Region Timber Harvesting and Processing Employment Survey. Final Report, December 15th, 2008. Vancouver, BC. Available online at: <http://www.for.gov.bc.ca/het/fibre.htm>

²⁹ Lefebvre (2008) reported an employment coefficient of 0.655 PY/1000m³ for primary processing in the region, and noted it was difficult to isolate a specific TSA coefficient due to the amount of log trading undertaken, particularly between the Kootenay Lake, Arrow, and Columbia TSA's. In order to look more specifically at employment within the project zone, we have estimated Kootenay specific employment by extrapolating data from the BC Fiber Flow reporting (BC Ministry of Forests, 2009), which used scaling data showed 50.4% of the Kootenay Lake harvest volume is processed within the Kootenay Lake TSA: Primary processing coefficient (Lefebvre, 2008) 0.655 PY/1000m³ * 50.4% (BC MOF, 2009) = 0.327 PY/1000m³

noted earlier, this is extremely conservative, as it assumes that the project zone forest industry would have directly and linearly expanded for each m3 in the baseline scenario, when in reality much of this volume would very likely have been substituted for lower margin log sources that would have been deferred (potentially until the Darkwoods baseline harvest declined).

However, it is also important to note, as shown in the CBB PDD, that the baseline scenario will drop harvest levels significantly after year 15. On average over the project, the baseline is expected to generate 64 person years per year valued at \$2.74 million per year, which is comparable to the project benefits in the monitoring period, which should be representative of ongoing levels for the project scenario in the long run.

Table 18. Direct economic benefits in this monitoring period, by scenario

Project Year	Project Harvest Volume (m3/yr)	Employment Generated (PY) Coefficient = 0.669	Employment Income (\$) Avg. \$42,630/yr
2013	261	0.2	\$7,444
2014	-	-	\$-
2015	12,077	8.1	\$344,430
2016	5,486	3.7	\$156,458
Avg/year	4,456	3.0	\$127,083
Total	17,824	11.9 ³⁰	\$508,331

Project Year	Baseline Harvest Volume (m3/yr)	Employment Generated (PY) Coefficient = 0.669	Employment Income (\$) Avg. \$42,630/yr
2013	301,285	201.6	\$8,592,489
2014	300,673	201.2	\$8,575,035
2015	301,390	201.6	\$8,595,483
2016	307,123	205.5	\$8,758,985
Avg/year	302,618	202.5	\$8,630,498
Total	1,210,471	809.8	\$34,521,991

³⁰ Note total project monitoring period employment includes this value plus the value shown in Table 19.

Table 19. Additional economic benefits in the project scenario (non-harvesting)

	Total \$	\$/year	Person Year (@\$42,630/yr)	Total Person Years (2013-2016)
NCC Staff Hours	\$885,838	\$221,460	5.2	20.8
DW Spending ³¹	\$1,035,162	\$258,791	6.1	24.3
Total	\$1,921,000	\$480,250	11.3	45.1

2. Community Impact Monitoring Activity 2: Climate Services Impact:

The project has been implemented as planned. Project ex-post harvest levels fluctuated, and overall were lower than planned in the PDD on an ex-ante basis, which provided more climate services benefit than planned. The project also experienced two forest fires which resulted in relatively low levels of emissions that decreased the expected climate benefits.

Overall, as shown in Table 20, the project produced significant emissions in the baseline scenario, and the implementation of the project generated substantial net climate services benefits for the community. Using an average of carbon market price indicators:

The baseline generated a negative impact to communities of -\$5.13 million per year, on average; or a total of -\$20.52 million over the monitoring period.

The project scenario generated a positive climate services impact of \$2.14 million per year on average, or a total of \$8.55 million over the monitoring period.

Collectively, implementing the project scenario resulted in a net positive community benefit for climate services of \$7.27 million per year on average, and a total of \$20.06 million over the monitoring period.

Note that these climate services values both represent a 'market value' if these net climate benefits needed to be purchased by the community, but also represent a proxy for the negative impact of climate change and adaptation might cost the communities over time for these emission levels³².

³¹ These include totals from NCC accounting records and relate to expenses such as forest research, management planning, carbon project costs, road deactivation and other expenses generally unique to the project scenario.

³² Note we are likely being very conservative using current market price indicators – the U.S. Government estimated the 'social cost of carbon' to be USD\$36.00/tCO₂e for federal agencies to weigh the costs of carbon emissions now against potential future damages from climate change (<https://obamawhitehouse.archives.gov/blog/2015/07/02/estimating-benefits-carbon-dioxide-emissions-reductions>).

Table 20. Darkwoods climate impact value by scenario

Net Emission Reductions by Scenario:

	Baseline (tCO ₂ e/year)	Project (tCO ₂ e/year)	Net Change (tCO ₂ e/year)
2013	(241,227)	130,376	371,602
2014	(285,826)	119,893	405,719
2015	(316,975)	50,274	367,248
2016	(257,335)	157,455	414,790
Period Total	(1,101,363)	457,977	1,559,360
Period Average per Yr	(275,341)	114,499	389,840

Carbon Pricing Scenarios (\$CDN):

	BC Legislation	Voluntary	CA ARB
Avg. \$CDN/tCO ₂ e	\$30.00	\$11.34	\$14.63

Climate Services Value Estimate Scenarios (Avg. \$CDN/Year)

	Carbon Price (\$CDN/tCO ₂ e)	Baseline Net Emissions (\$/year)	Project Net Emission Reductions (\$/year)	Net Climate Benefit (\$/year)
BC Carbon Tax Scenario	\$30.00	\$(8,247,315)	\$3,435,173	\$11,682,488
Voluntary Market Scenario	\$11.34	\$(3,117,485)	\$1,298,495	\$4,415,980
CA ARB Scenario	\$14.63	\$(4,022,112)	\$1,675,291	\$5,697,403
Avg. Value Impact (\$/yr):		\$(5,128,971)	\$2,136,319	\$7,265,290

Climate Services Value Estimate Scenarios (Total Period \$CDN)

	Carbon Price (\$CDN/tCO ₂ e)	Baseline Net Emissions (total \$)	Project Net Emission Reductions (total \$)	Net Climate Benefit (total \$)
BC Carbon Tax Scenario	\$30.00	\$(32,989,260)	\$13,740,690	\$46,729,950
Voluntary Market Scenario	\$11.34	\$(12,469,940)	\$5,193,981	\$17,663,921
CA ARB Scenario	\$14.63	\$(16,088,450)	\$6,701,163	\$22,789,612
Avg. Value Impact:		\$(20,515,883)	\$8,545,278	\$29,061,161

3. Community Impact Monitoring Activity 3 – Other Ecosystem Services Provisioning Impact

The project has been implemented as planned, and there have been no unplanned land use changes to the project area during this monitoring period that materially affect ecosystem services provisioning.

The qualitative assessment non-climate ecosystem services provisioning impacts has been updated for the monitoring period, as shown in Table 21. There is no evidence of material changes to any of these assessment values during this monitoring period.

Table 21 - Qualitative Assessment of Non-Climate Ecosystem Services Provision Impacts.

Ecosystem Service	Baseline Impacts	Project Impacts	Magnitude of Value
Water Quality/Quantity	-	++	Significant ³³
Key Species Habitat	--	++	Significant ³⁴
Pollinator Services	-	+	Low-Moderate ³⁵
Recreational Hunting/Fishing	+ ³⁶	++ ³⁷	Low to Significant
Outdoor Recreation	-	+	Moderate ³⁸
Wetland Protection	-	+	Low ³⁹

³³ In particular, the Darkwoods property contributes significantly to the valuable Kootenay Lake fishery, worth \$5-10 million per year to the local community (page 8: http://www.env.gov.bc.ca/kootenay/fsh/main/pdf/KLAP%20Kootenay%20Lake%20Action%20Plan%20final%209_May_2016.pdf).

³⁴ In particular, the endangered mountain caribou, grizzly bears, wolverines and other key species with extensive regional planning, community involvement, and public costs. The Darkwoods project significantly contributes habitat, migration corridors, and other values to these and other species. This contributes to the success of these regional efforts and directly reduces short and long term community costs related to research, monitoring, habitat protection, and public land allocation/acquisitions.

³⁵ As discussed in the CBB PDD, fully functional ecosystems in the project scenario likely support native pollinators, which may have some positive impact on local Creston Valley agriculture, either through direct pollination services or more likely contributes to species diversification, stabilizing closer populations, and providing replacement populations as necessary.

³⁶ Some key game species (i.e. deer) may benefit from improved browse habitat created by logging in the baseline scenario, although extensive logging may reduce other habitat attributes for such species. It is unclear whether access for hunting in the baseline scenario would have been allowed, as regional property practices are mixed. To be conservative, a positive impact rating is given.

³⁷ The project would also benefit key game species such as deer and elk (browse in harvested areas, winter cover in retained forest, predation cover, reduced road access, etc.). The benefit to recreational fishery on Kootenay Lake may be significant, as undisturbed Darkwoods stream systems contribute directly to water quality, quantity, temperature, invertebrate populations, and fish hatchling populations into Kootenay Lake and other watersheds. NCC has allowed public hunting on parts of the property (whereas the previous owner excluded hunting) which is an additional benefit to local community.

³⁸ As shown in Table 22, Darkwoods provides direct benefits to 200-350 backcountry recreational users of the property each year. It is doubtful the fully operational and heavily logged property in the baseline scenario would provide a similar recreational draw.

³⁹ Significant wetlands are generally protected by minimum legal operating practices, and hence would have minor impacts from either scenario; however the drainages into and out of wetlands and small wetlands may be impacted by logging in the baseline scenario, leading to a negative overall impact rating.

4. Community Monitoring Activity 4 – Community Amenity Value Impacts

The project has been implemented as planned, and there have been no unplanned land use changes to the project area during this monitoring period. NCC continues to be engaged with a significant number of community groups and users related to the project area and project activities.

No new quality of life or community environmental values surveys have been located since the detailed materials reviewed in the CCB PDD. However, anecdotally, the Nelson and Creston area continues to be a highly diversified community with highly desirable quality of life features. There is also no evidence of any material shift in community values since the community surveys described in the PDD. Therefore, we assess that the project fully supported community amenity values during this monitoring period.

For example, the Darkwoods project area supported non-motorized recreational and backcountry use via a permit system, continued to support hunting activities, supported scientific research and projects of value to the region and communities, and continued to provide visual benefits to the Creston area, etc.

Table 22 Darkwoods public access permit monitoring results

Darkwoods Public Access Permits	Year	# Permits	Est. Non-Permitted Use
Public Access	2013	172	150
Public Access	2014	98	150
Public Access	2015	80	150
Public Access	2016	138	150

Table 23. Recreational groups engaged with NCC during the monitoring period

Recreational Groups	Years
Kootenay Mountaineering Club	2013-2016
Beaver Mountain Snowmobile Association	2013-2016
Kokanee Snowmobile Club	2013-2016
Nelson Snowgoers Club	2013-2016
Creston Rod and Gun Club	2013-2016
Association of BC Snowmobile Clubs	2013-2016
West Kootenay Naturalist Club	2013
West Kootenay Camera Club	2013
Selkirk Rock and Mineral Club	2015
Western Canada Bryophyte and Lichen Interest Group	2014
Nelson Camera and Naturalist Club	2013
Climbers Association	2013

Table 24. Scientific researchers engaged on Darkwoods during the monitoring period

Scientific Researchers	Topic	Years
Michael Proctor	Grizzly Bears	2013-2016
Adrian Leslie	Whitebark pine	2013-2014
Marc-Andre Beaucher	Nighthawks	2015
Grant McHutchen	Grizzly Bears	2013-2016
Masse Environmental Consultants	Bull Trout Redd Surveys	2015
Stefan Himmer	Huckleberries	2015
Gerry Nellestijn	Bull Trout	2016
Olivia Lee	Bryophytes and Lichen	2014
Leo DeGroot	Caribou	2013-2016
Highlander Forestry	FLNRO subcontractors	2014
Cori Lauson	Bats	2014
Jeremy deWaard	Anthropods	2014
Pam Dykstra	FLNRO subcontractors	2013
Olivia Lee	botany	2014
Steve Ardnt	kokanee and Bull Trout	2013-2016
BC Conservation Data Centre	red and blue listed species	2016

4.3.3 Dissemination of Monitoring Plan and Results (CM3.3)

The primary form of general project information distribution is through the NCC project webpages. Project implementation and project area activities are also communicated through meetings with key stakeholders. See additional description in Section 5.3.3, and a non-exhaustive list of stakeholder meetings during the monitoring period is shown in Section 5.3.3, Table 23, and Table 24.

4.4 Optional Criterion: Exceptional Community Benefits

Not Applicable.

4.4.1 Barriers to Benefits (GL2.3)

Not Applicable.

4.4.2 Protections for Poorer and More Vulnerable Households and Individuals (GL2.4)

Not Applicable

5 BIODIVERSITY

5.1 Net Positive Biodiversity Impacts

5.1.1 Biodiversity Changes (B1.1)

As per the Darkwoods CCBA PDD, the management of biodiversity is outlined in the Darkwoods Conservation Area Property Management Plan 2011-2016 (NCC, 2011)⁴⁰ for this monitoring period. The NCC (2011) document provides detailed descriptions of the biodiversity effectiveness program, including background, references and assumptions. Consistent with this plan, biodiversity effectiveness monitoring is undertaken on an annual or bi-annual basis to track progress towards property management goals.

As shown in Table Table 25^{41,42}, the project area has maintained or improved the key biodiversity effectiveness monitoring indicators during this monitoring period. The project has improved several factors since the project began, and the project is making good progress towards the desired future conditions.

Table 25 - Biodiversity Targets, Monitoring Data, and Trends. 2011-2015 Monitoring Period

Target	Poor	Fair	Good	Very Good	Methods	Source of Data
Dry interior cedar-hemlock forest						
Proportion of old forest in ICHdw	<0.1	0.1-0.15	0.15-0.21	>0.21		
2011 assessment (trend: increasing)	0.02					
2013 (trend: increasing)		0.09			Total forested ICHdw1 = 4328.5 ha, Total ICHdw1 > 140 yrs = 397 ha OR 9% **Forested =	Used Ecora VRI, Deb MacKillop BGC

⁴⁰ This plan was updated in 2016 (NCC Darkwoods Property Management Plan 2016-2021 (NCC, 2016), which will guide biodiversity management in future monitoring periods.

⁴¹ Note this table summarizes the trends up to end of 2015 – trends in 2016 are consistent with 2015, and hence this table is representative of the results for the entire monitoring period.

⁴² This table is consistent with the biodiversity targets, measures, methods, and data sources of (the somewhat blurry) Table 23 in Section of the Darkwoods CCBA PDD. A clear version of this table is available.

Target	Poor	Fair	Good	Very Good	Methods	Source of Data
					Spc1 not blank, remove age < 3 (NSR)	
2015 (trend: increasing)		0.09			Total forested ICHdw1 = 4328.5 ha, Total ICHdw1 > 140 yrs = 397 ha OR 9% **Forested = Spc1 not blank, remove age < 3 (NSR). No harvest or fires within this defined area in 2014/15	Used Ecora VRI, Deb MacKillop BGC
Desired future condition				0.21		
Grizzly bear						
Proportion of core in SS 3	0-0.1	0.1-0.15	0.15-0.18	>0.19		
2011 assessment (trend: decreasing)				0.25		
2013 (trend: decreasing)				0.24	strstg3 = 6244/26211 = 24%or Core is SS3	Used Ecora Struct Stg
2015 (trend: decreasing)				0.24	strstg3 = 6244/26211 = 24%or Core is SS3	Used Ecora Struct Stg
Desired future condition				>0.19		
Proportion of core no access/non-motorized	0-0.35	0.25-0.5	0.5-0.75	0.75-1		
2010 assessment			0.67			

Target	Poor	Fair	Good	Very Good	Methods	Source of Data
(trend: increasing)						
2013 (trend: stable)				0.87	Of the 26,211 ha of core mapped habitat, 22830 ha is zoned No Access or 87% of Core is No Access	New Darkwoods boundary, Primary Zone expanded
2015 (trend: stable)				0.87	Of the 26211 ha of core, 22830 ha is zoned No Access or 87% of Core is No Access.	New Darkwoods boundary, Primary Zone expanded
Desired future condition			0.67			
Mountain caribou ⁴³						
Proportion of capable habitat now suitable	<0.4	0.4-0.6	0.6-0.8	0.8-1.0		
2010 assessment (trend: increasing)		0.51				
2013 (trend: increasing)		0.51				
2015 (trend: increasing)		0.51				
Desired future condition			0.6			
Hydro-riparian ecosystems						

⁴³ Also note, Table 23 in the CCB PDD includes reference to monitoring % of suitable caribou habitat that was disturbed by natural or anthropological means. Although project effectiveness monitoring reporting focuses on project activities impacts on caribou habitat (zero impact; see Table 25 and Figure 6), additional GIS analysis shows a total caribou habitat disturbance of 75 ha (due to one wildfire in 2014 that occurred within the critical caribou habitat area), which amounts to a total caribou habitat impact of 0.334% during this monitoring period.

Target	Poor	Fair	Good	Very Good	Methods	Source of Data
Barriers to fish flow/passage	>5	2-5	1-2	0		
2010 assessment (trend: stable)				0		
2013 (trend: stable)				0	7 wood culverts removed (does not include removal of steel)	Staff review of activities
2015 (trend: stable)				0		
Desired future condition				0		
Proportion of watersheds with Equivalent Clearcut Areas of <20%	<0.5	0.5-0.75	0.75-1	1		
2010 assessment (trend: increasing)		0.6				
2013 (trend: stable)				1	See attached worksheet	Used carbon VRI resultant
2015 (trend: stable)				1	See attached worksheet	Used carbon VRI resultant
Desired future condition				1		
Number of bridges on the property at high risk of failure	>5	3-5	1-2	0		
2010 assessment (trend: increasing)				0		
2013 (trend: stable)				0	5 Bridges removed since 2009, 1 installed	Staff review of activities

Target	Poor	Fair	Good	Very Good	Methods	Source of Data
2015 (trend: stable)				0	6 Bridges removed since 2009, 1 installed	Staff review of activities
Desired future condition				0		
Kilometres of roads rated at medium- or high risk of failure	>50	25-50	1-25	0		
2010 assessment (trend: increasing)		25-50				
2013 (trend: stable)				0	Up to 2013 - 147 km deactivated out of 603 total km of road	Staff review of activities
2015 (trend: stable)				0	Up to Fall 2015 - 218 km deactivated out of 603 total km of road	Staff review of activities
Desired future condition				0		
Old-growth cedar-hemlock forest						
Proportion old forest in ICH	<0.1	0.1-0.15	0.15-0.21	>0.21		
2010 assessment (trend: increasing)		0.12				
2013 (trend: increasing)				0.28	Total forested ICHmw4 = 5638.3 ha, Total ICHmw4 > 140 yrs = 1601 ha OR 28% **Forested = Spc1 not blank, remove age < 3 (NSR)	Used Ecora VRI, Deb MacKillop BGC

Target	Poor	Fair	Good	Very Good	Methods	Source of Data
2015 (trend: increasing)				0.28	Total forested ICHmw4 = 5638.3 ha, Total ICHmw4 > 140 yrs = 1601 ha OR 28% **Forested = Spc1 not blank, remove age < 3 (NSR). No harvest or fires within this defined area in 2014/15	Used Ecora VRI, Deb MacKillop BGC
Desired future condition				0.21		
Rare ecosystems and features						
Proportion of sites protected	<0.5	0.5-0.75	0.75-1	1		
2010 assessment (trend: stable)				1		
2013 (trend: stable)				1		
2015 (trend: stable)						
Desired future condition				1		

*all indicator results are based on GIS analysis conducted by NCC (Kathleen McGuinness)

5.1.2 High Conservation Value Protection (B1.2)

As a conservation-based project, project activities are focused away from high value conservation areas on the property, or planned in a way to minimize impacts to key biodiversity targets. During the monitoring period, NCC has continued to mitigate potential impacts to high conservation values by:

Protecting and/or managing old growth forest areas on the property, and otherwise managing project activities away from important conservation values;

Maintaining access control on the property through gates, access permits, and road deactivation;

Continuing to explore opportunities for key habitat restoration/management (no material direct habitat management actions were undertaken during this monitoring period);

Continuing to identify and eliminate problematic invasive species on the property (see Section 5.1.3).

The Darkwoods CCB PDD (Section G1.8) generally discusses the potential high conservation values on the project area, including: landscape level ecosystems (old growth forests), threatened or rare ecosystems (hydro-riparian ecosystems; dry interior Cedar-Hemlock (ICH) forests), critical ecosystem services (hydro-riparian ecosystems, caribou, native pollinators), local community needs, and traditional use.

As per the PDD, local community needs and traditional needs are not materially present or impacted by the project.

During this monitoring period, the project has not materially impacted the remaining potential high conservation values; in particular: old growth forests, ICH forests, hydro-riparian areas, and caribou habitat⁴⁴. As shown in Figure 3, Figure 4, 5, and Figure 6, the project activities in this monitoring period did not impact any old growth areas, hydro-riparian areas, ICH forests, or caribou critical habitat during this monitoring period.

⁴⁴ Other critical species mentioned in the CCB PD include grizzly, wolverines, and native pollinators. Grizzlies and wolverines are wide ranging species that are more impacted by human road usage and landscape scale ecosystem protections (i.e. the old growth areas). Project logging does not impact these species specifically other than by proxy of minimizing widespread road usage and protecting larger scale areas of natural ecosystems. Native pollinators are wide spread and also related to the general protection of larger areas of natural ecosystems, and hence are not impact by the small project harvest areas and are represented by the old growth area mapping.

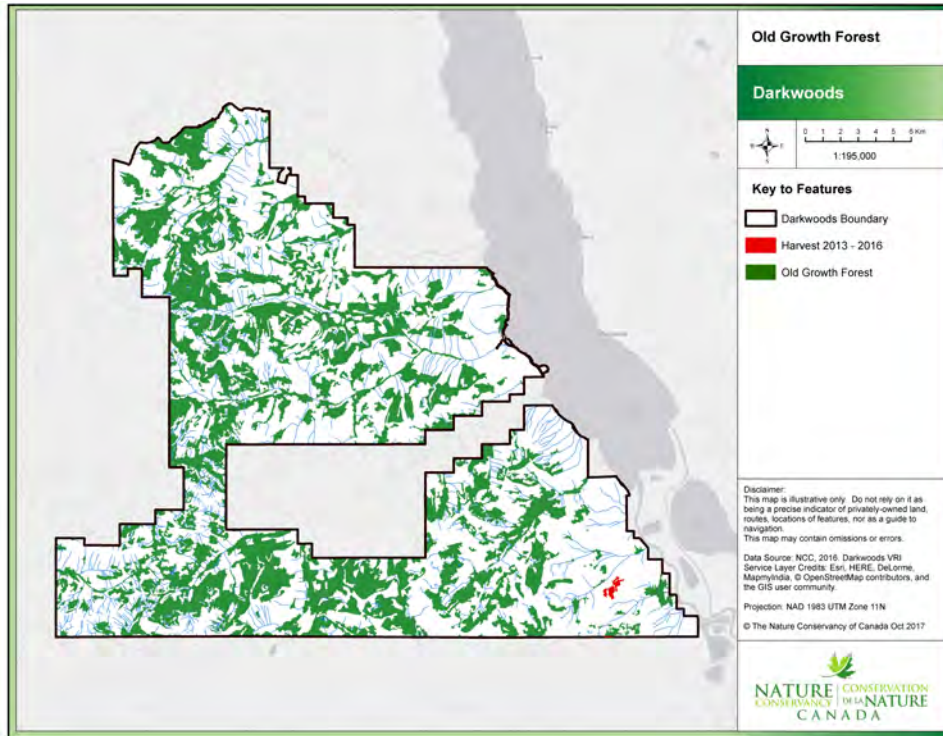


Figure 3 - 2013-2016 Project Activities vs. Old Growth Forest Areas

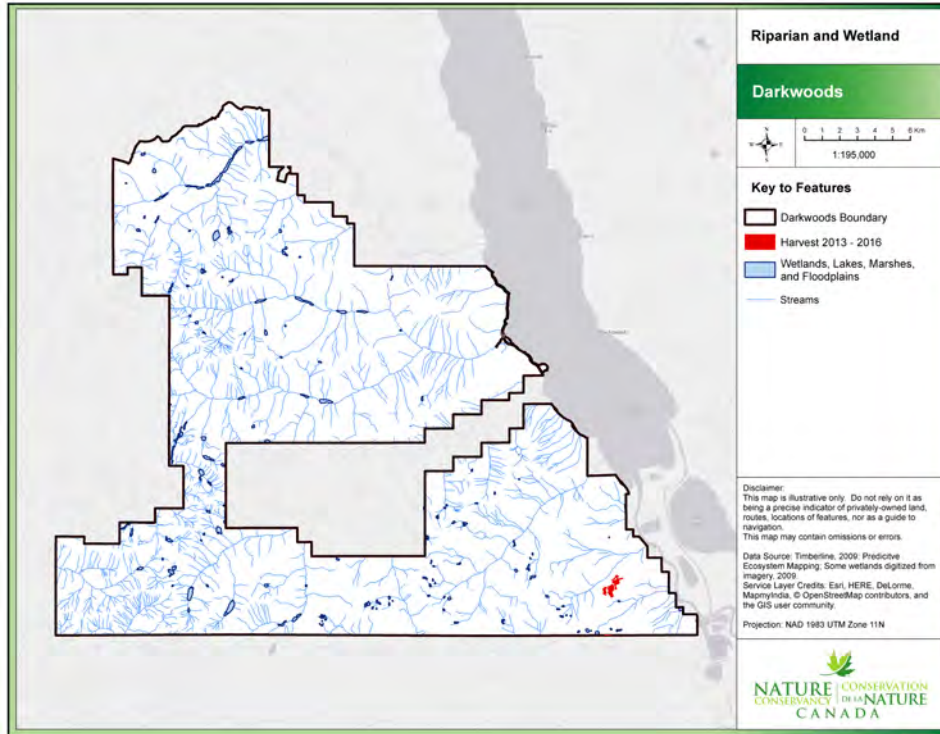


Figure 4. 2013-2016 Project Activities vs. Hydro-Riparian Areas

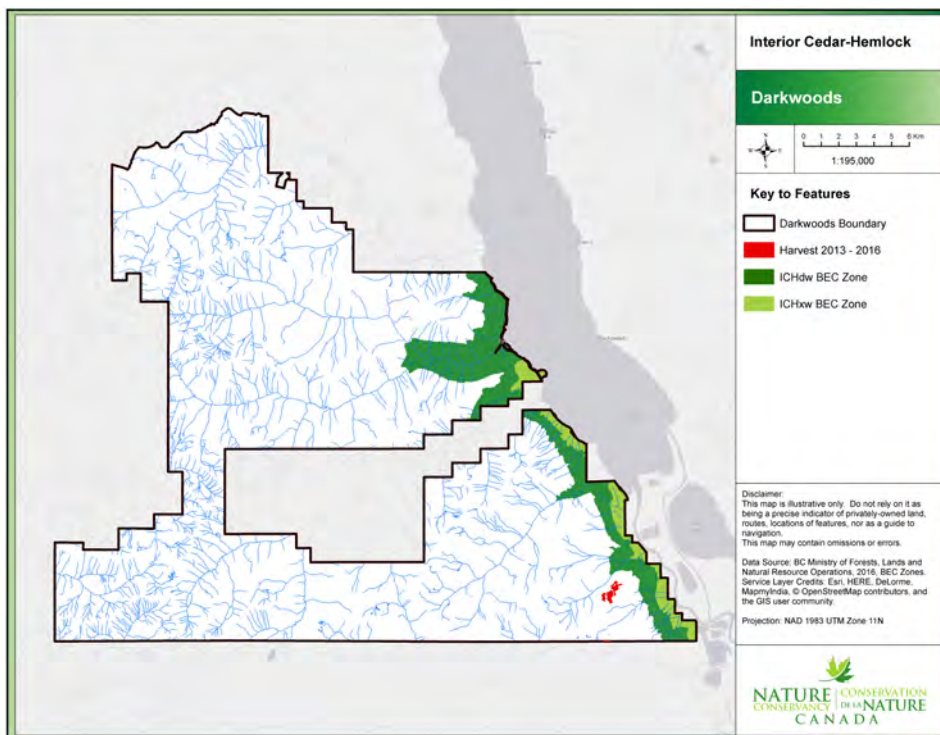


Figure 5. 2013-2016 Project Activities vs. ICH Forests

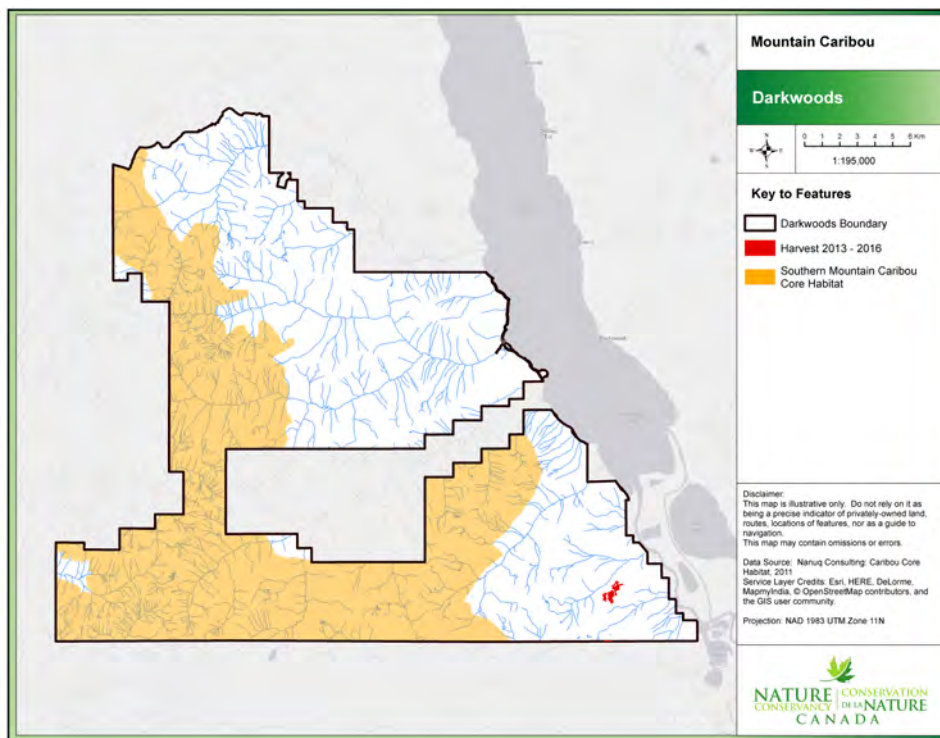


Figure 6. 2013-2016 Project Areas vs. Caribou Critical Habitat

5.1.3 Invasive Species (B1.3)

No known invasive species have been introduced to Darkwoods as a result of project activities during the monitoring period. Invasive species occurrence data are available from the Invasive Alien Plant Program (IAPP) (<https://catalogue.data.gov.bc.ca/dataset/invasive-alien-plant-site>), run by the BC Government's Ministry of Forests, Lands and Natural Resource Operations (FLNRO). The IAPP database contains invasive plant surveys, treatments, and activity plans for the entire province of BC. The comprehensive data in IAPP is entered by a wide variety of user groups (ministries, regional districts, weed committees, forest licensees, utilities, conservation groups, federal departments and others) on an on-going basis. A total of 8 species were recorded during the monitoring period on the project area, all of which were also documented prior to the start of the monitoring period in 2013, and hence the number of invasive species has not increased during this monitoring period. Further, NCC maintains an active invasive species removal program. Invasive species almost exclusively occur along the main roads where public vehicles are present frequently. NCC monitors the presence of invasive species on a periodic basis across the property (not just the main roads, although that is where they are most likely to occur) and through informal observation and reporting by NCC staff and researchers using the property. Then, all known invasive species are removed on a periodic or annual basis by chemical, biological, or mechanical means. Figure 7 shows the occurrence of invasive species removal activities during this monitoring period. As NCC maintains this active removal program, it can be assumed that the population of invasive species is not increasing (as any known are removed).

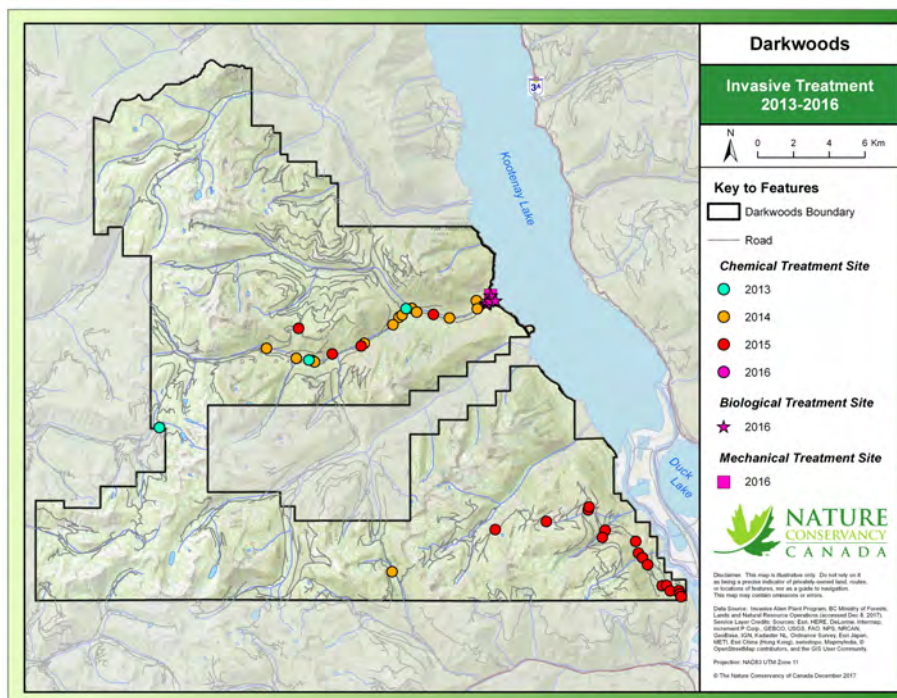


Figure 7. Darkwoods Invasive Species Removal Program 2013-16

5.1.4 Impacts of Non-native Species (B1.4)

Not Applicable. Non-native species are not used during project activities.

5.1.5 GMO Exclusion (B1.5)

NCC guarantees that no GMOs were used on the Darkwoods property to generate GHG emissions reductions or removals. All of the seedlings planted during the monitoring period were grown by PRT – Harrop nursery <https://www.prt.com/contact/nursery-locations/prt-harrop>. PRT receives all their seeds from the Tree Seed Center of the Tree Improvement Branch in the BC Ministry of Forests, Lands and Natural Resource Operations (FLNRO). From the FLNRO Tree Improvement Branch website: (at bottom of webpage <https://www2.gov.bc.ca/gov/content/industry/forestry/managing-our-forest-resources/tree-seed/seed-production>): “Tree Improvement Branch is not involved in any genetic engineering, gene insertion, gene splicing or biotechnology that results in genetically modified organisms. In keeping with the Government’s broad stewardship responsibilities for Crown land, Tree Improvement Branch has ensured that no genetically modified tree seed has been registered or used in operational forest planting on Crown land in British Columbia.”

5.2 Offsite Biodiversity Impacts

5.2.1 Negative Offsite Biodiversity Impact Mitigation (B2.2)

Not Applicable - It is exceedingly unlikely that the project will generate negative offsite biodiversity impacts. The principal objectives of the project are to promote biodiversity habitat and values. The size of the project area (55,000 ha) is such that offsite biodiversity impacts are likely to be positive rather than negative, particularly given the extensive connectivity provided by the project area to other protected landscapes.

5.2.2 Net Offsite Biodiversity Benefits (B2.3)

The Darkwoods property is a site of global significance for biodiversity conservation. With a total area of 55,200 ha, the size of the property is significant. The current and long-term biodiversity potential is very high. To date, 200 animal and 219 plant species have been confirmed on the property (Steeger and Machmer 2009). Ninety-seven site series (ecosystems) have been mapped for the property (NCC 2010) for the baseline inventory.

IUCN red-list species found on the property to date are: Grizzly Bear (*Ursus arctos*)(Threatened), Wolverine (*Gulo gulo*)(Near Threatened), Bull Trout (*Salvelinus confluentus*) (Vulnerable). Other notable species: In total, 19 Globally, Nationally, and/or Provincially significant species-at-risk are predicted to occur on the property, 15 of which have been confirmed (Steeger and Machmer 2009).

Darkwoods likely provides significant habitat for at least 7 of these species-at-risk, including Grizzly Bear (Special Concern; blue list), Mountain Caribou (*Rangifer tarandus caribou*) (Threatened; Red list), Olive-sided Flycatcher (*Contopus cooperi*) (Threatened; Blue list), Peregrine Falcon (*Falco peregrinus*) (Special Concern; Red list), Western Toad (*Bufo boreas*) (Special Concern; Yellow list), Wolverine (Special Concern; Blue list) and Bull Trout (Blue list).

The biodiversity significance of Darkwoods is partly a consequence of its geographic extent but is also related to its topographical gradient. From an elevational perspective, the property extends from warm, moist valley bottoms, to cold and dry, high elevation Engelmann spruce/Subalpine fir forests. Under the warmer climate predicted for the region, many lower elevation species are likely able to migrate upwards as their original habitat deteriorates while upslope conditions begin to ameliorate.

In addition to the magnitude of the area encompassed by Darkwoods, the property has considerable importance because of its adjacency to, and connectivity with, three provincial protected areas. These are West Arm Provincial Park (WAPP), Midge Creek Wildlife Management Area (MCWMA), and the Creston Valley Wildlife Management Area (a 7,000 ha RAMSAR site; see s. G1.8). WAPP is an area of about 25,300 ha that extends along the shore of Kootenay Lake from Nelson to Harrop, BC and up to the peaks behind these towns. As the natural habitat for several endangered species, the area was protected for its biodiversity value and which also serves to protect Nelson's water source. The MCWMA is a 15,163-hectare protected area that provides critical lower elevation wildlife habitat for a variety of species-at-risk, including grizzly bear and mountain caribou. In totality, Darkwoods constitutes about 50% by area of a contiguous protected area of about 100,000 ha.

5.3 Biodiversity Impact Monitoring

5.3.1 Biodiversity Monitoring Plan Development (B3.3)

Not applicable.

5.3.2 Biodiversity Monitoring Results (B3.1, B3.2)

NCC undertook annual biodiversity monitoring during this monitoring period, which are compiled early in the subsequent year by NCC staff and GIS personnel. The monitoring results of available during this monitoring period are captured in the following files: Darkwoods 2013 Effectiveness Monitoring Report.xlsx; Darkwoods 2014 Effectiveness Monitoring Report.xlsx; Darkwoods 2015 Effectiveness Monitoring Report.xlsx; and Darkwoods 2016 Effectiveness Monitoring Report.xlsx. These reports are summarized in Table 26. Darkwoods tracks and reports biodiversity effectiveness monitoring results on a backward looking annual calendar year basis – the dates, frequencies and locations of specific monitoring are not recorded, and are variable and may be undertaken on whatever schedule is efficient for field work and data analysis in order to report on the previous year, using the most up to date GIS data and other sources as applicable at the time each monitoring report is produced.

Summarized, the project has been able to keep maintain or improve key biodiversity targets stable or improving over this monitoring period. Project activities have been focused outside of key biodiversity target areas. See also Table 25 for data on biodiversity effectiveness trends and targets.

Table 26 - Summarized Annual Biodiversity Effectiveness Monitoring Results

Target	2013	2014	2015	2016	Methods	Source
Dry interior cedar-hemlock forest						
Proportion of old forest in ICHdw	0.09	0.09	0.09	0.09	Total forested ICHdw1 = 4328.5 ha, Total ICHdw1 > 140 yrs = 397 ha OR 9% **Forested = Spc1 not blank, remove age < 3 (NSR). No harvest within this defined area in this period.	Used Ecora VRI, Deb MacKillop BGC, GIS harvest data.
Grizzly bear						
Proportion of core in Structural Stage 3 (SS3)	0.24	0.24	0.24	0.24	strstg3 = 6244/26211 = 24% or Core is SS3	Used Ecora VRI Struct Stg
Proportion of core no access/non-motorized	0.87	0.87	0.87	0.87	Of the 26,211 ha of core mapped habitat, 22830 ha is zoned No Access or 87% of Core is No Access	Darkwoods boundary, Primary Zone expanded
Mountain caribou ⁴⁵						
Proportion of capable habitat lost to disturbance	0.00	0.00	0.00	0.00	No logging in caribou habitat areas in the monitoring period	GIS annual harvest data
Hydro-riparian ecosystems						
Barriers to fish flow/passag e	0.00	0.00	0.00	0.00	No new activity in period	Staff reporting of activities
Proportion of watersheds with	1.00	1.00	1.00	1.00	See annual EM report worksheets	Used carbon VRI resultant

⁴⁵ Also see: Footnote 43.

Equivalent Clearcut Areas of <20%						
Number of bridges on the property at high risk of failure	0.00	0.00	0.00	0.00	6 Bridges removed since 2009, 1 installed	Staff reporting of activities
Kilometres of roads rated at medium- or high risk of failure	0.00	0.00	0.00	0.00	Up to Fall 2015 - 218 km deactivated out of 603 total km of road	Staff reporting of activities
Old-growth cedar-hemlock forest						
Proportion old forest in ICH	0.28	0.28	0.28	0.28	Total forested ICHmw4 = 5638.3 ha, Total ICHmw4 > 140 yrs = 1601 ha OR 28% **Forested = Spc1 not blank, remove age < 3 (NSR). No harvest or fires within this defined area in period.	Used Ecora VRI, Deb MacKillop BGC, GIS harvest data.
Rare ecosystems and features						
Proportion of sites protected	1.00	1.00	1.00	1.00	No new activity in period	Staff reporting of activities

5.3.3 Monitoring Plan and Results Dissemination (B3.3)

NCC uses a variety of targeted methods to disseminate project information and monitoring results. NCC regularly publishes stories on its website and blog about the natural and conservation features of Darkwoods in order to provide information and share compelling conservation stories with the general public. At least two of the approximately 12 stories published between 2013 and 2016 directly addressed the Forest Carbon Project. The remaining stories or web updates covered topics such as mountain caribou, white-bark pine, grizzly bear and the use of technology in sharing the remote landscape with the broader public. Public talks featuring the grizzly bear research of Michael Proctor have been held a number of times in various communities during the reporting period.

More technical information related to monitoring results were shared directly to researchers and other interested stakeholder through in person meetings and direct communications. The table below is a non-

exhaustive list of meetings used to disseminate biodiversity monitoring results and other project information:

Date	Staff Presenter(s)	Event	Topic	Location	Audience	# of Attendees
March 2013	Nancy Newhouse	Annual Darkwoods Researchers meeting	All biodiversity monitoring	Nelson	Researchers from government, non-profit and independent	15
March 2014	Nancy/Hillary	Annual Darkwoods Researchers meeting	All biodiversity monitoring	Nelson	Researchers from government, non-profit and independent	15
22-Jan-2014	Michael Proctor	Nature Talks	Grizzly Bears	Creston	General public	40
March 2015	Adrian Leslie	Annual Darkwoods Researchers meeting	All biodiversity monitoring	Nelson	Researchers from government, non-profit and independent	15
01-Jul-2015	Adrian Leslie	Field tour	Darkwoods wildlife habitat	Darkwoods	Canadian Wildlife Service employees (retired)	23
18-Nov-2015	Adrian Leslie	Nature Talks	Darkwoods whitebark pine	Victoria (Oak Bay High)	NCC supporters, general public	75
01-Dec-2015	Adrian Leslie	Kootenay Conservation Program committee meeting	Darkwoods biodiversity planning	Nelson	KCP Stewardship committee members	20
07-Dec-2015	Adrian Leslie	Beaver Mountain Snowmobile Club meeting	mountain caribou	Fruitvale	club members (snowmobilers)	20
18-Feb-2016	Adrian Leslie	Wildsight meeting/event	Darkwoods biodiversity	Creston	General Public	45
March 2016	Adrian Leslie	Annual Darkwoods Researchers meeting	All biodiversity monitoring	Nelson	Researchers from government, non-profit and independent	15
25-Nov-2016	Adrian Leslie	Kootenay Conservation Program event	Tye restoration project	Creston	KCP members	28

6 ADDITIONAL PROJECT IMPLEMENTATION INFORMATION

Not Applicable.

7 ADDITIONAL PROJECT IMPACT INFORMATION

For reference (excerpt from the Darkwoods CCBA PDD):

GL3. Exceptional Biodiversity Benefits

The Darkwoods property is a site of global significance for biodiversity conservation. With a total area of 55,200 ha, the size of the property is significant. The current and long-term biodiversity potential is very high. To date, 200 animal and 219 plant species have been confirmed on the property (Steeger and Machmer 2009). Ninety-seven site series (ecosystems) have been mapped for the property (NCC 2010) for the baseline inventory.

IUCN red-list species found on the property to date are: Grizzly Bear (*Ursus arctos*)(Threatened), Wolverine (*Gulo gulo*)(Near Threatened), Bull Trout (*Salvelinus confluentus*) (Vulnerable). Other notable species: In total, 19 Globally, Nationally, and/or Provincially significant species-at-risk are predicted to occur on the property, 15 of which have been confirmed (Steeger and Machmer 2009).

Darkwoods likely provides significant habitat for at least 7 of these species-at-risk, including Grizzly Bear (Special Concern; blue list), Mountain Caribou (*Rangifer tarandus caribou*) (Threatened; Red list), Olive-sided Flycatcher (*Contopus cooperi*) (Threatened; Blue list), Peregrine Falcon (*Falco peregrinus*) (Special Concern; Red list), Western Toad (*Bufo boreas*) (Special Concern; Yellow list), Wolverine (Special Concern; Blue list) and Bull Trout (Blue list).

The biodiversity significance of Darkwoods is partly a consequence of its geographic extent but is also related to its topographical gradient. From an elevational perspective, the property extends from warm, moist valley bottoms, to cold and dry, high elevation Engelmann spruce/Subalpine fir forests. Under the warmer climate predicted for the region, many lower elevation species are likely able to migrate upwards as their original habitat deteriorates while upslope conditions begin to ameliorate.

In addition to the magnitude of the area encompassed by Darkwoods, the property has considerable importance because of its adjacency to, and connectivity with, three provincial protected areas. These are West Arm Provincial Park (WAPP), Midge Creek Wildlife Management Area (MCWMA), and the Creston Valley Wildlife Management Area (a 7,000 ha RAMSAR site; see s. G1.8). WAPP is an area of about 25,300 ha that extends along the shore of Kootenay Lake from Nelson to Harrop, BC and up to the peaks behind these towns. As the natural habitat for several endangered species, the area was protected for its biodiversity value and which also serves to protect Nelson's water source. The MCWMA is a 15,163 hectare protected area that provides critical lower elevation wildlife habitat for a variety of species-at-risk, including grizzly bear and mountain caribou. In totality, Darkwoods constitutes about 50% by area of a contiguous protected area of about 100,000 ha.

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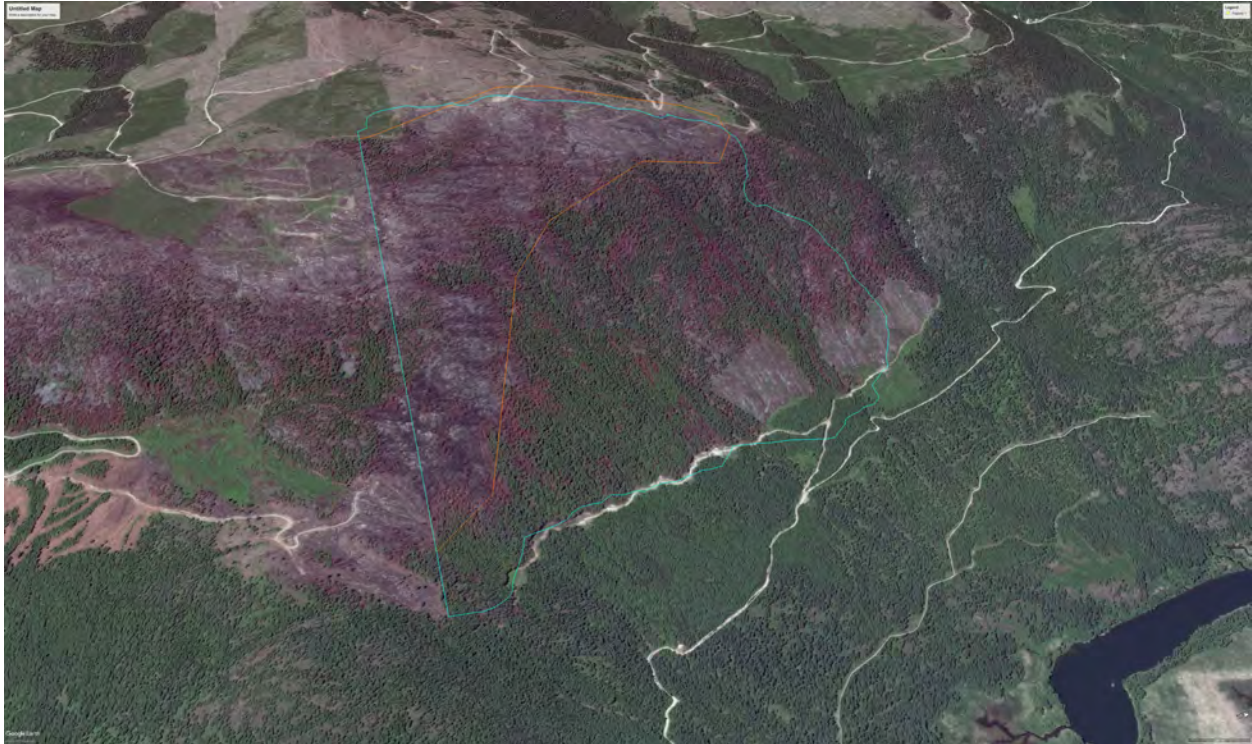
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8 APPENDIX 1: EVALUATION OF 2015 FIRE BOUNDARY ON SATELLITE IMAGE

Figure A1. Overlay 2015 fire boundaries on a satellite image from Google Earth taken in 2016. The Orange border shows the map of the 2015 stand-replacing fire boundary created by the Darkwoods staff during a helicopter fly over and the blue line is from the provincial fire database showing the whole area affected by the fire. The patches on the lower right side had already been harvested. The straight boundary on the left side represents the southern limit of the Darkwoods property.



9 APPENDIX 2 – VCS NON-PERMANENCE RISK ASSESSMENT

The VCS AFOLU Non-Permanence Risk Assessment is attached to this Monitoring Report in 2 files:

VCS Risk Report Calculation Tool, v3.1-DW 2013-2016

VCS Non-Permanence Risk Report (Short) - Darkwoods 2013-16 Verification v2

There has been no change to the Non-Permanence Risk Rating during this monitoring period. It remains at 10%.